ProbeFlow: Contracts, Derivations, Recorded Cases, and Development Evidence

Research Forge · 19 October 2026 · Layout v20 · Research content v20

Supplement to compact v20, 2026-10-19. New exposition of existing saved evidence; no new model, fitting, independent empirical study, native reproduction, or peer review. The inherited v19 manuscript below is preserved in full.

Abstract

Detailed assumptions, derivations, full comparison contracts, negative evidence and reproducibility notes accompany the compact main paper. The full previous scientific record is retained without changing its reported numbers or evidence scope. No new experiment, model invocation or independent sample is introduced by this document split.

1 S1. Typed observations and the execution boundary

Separate a record's existence from its disclosure. A private record contains sensor traces, labels, and cached predictions. A public observation contains only source means actually returned after validated requests. A locked policy specifies a fixed mask or a root, hexadecimal threshold, and two terminal masks. The executor owns affordability, acquired-source bookkeeping, and reveal events; the auditor owns scoring. These are distinct types of permission even when all objects reside in one trusted Python process.

The implemented decision signature is next_action(policy, observed, acquired). It receives no cycle identifier, target, undisclosed source value, or terminal prediction bank. A gate requests its root if absent; otherwise it uses only observed[root] and the immutable threshold to choose a terminal mask. The next not-yet-acquired source is requested in ascending source index, or STOP is returned. The executor checks no repeat and spent+price≤B before revealing a mean. The trace records the revealed mean as float.hex together with the charged source quote.

The terminal predictor may use the purchased source's predefined feature bundle even though branching uses only its mean. Access to that bundle is authorized by acquiring the source; it is not a license for the controller to inspect every hidden transform before choosing an action. This distinction supports inspection and poison-value tests of trusted code. It is not process isolation, a capability-secure service, or a guarantee against arbitrary malicious code.

Object / owner	Permitted contents	Excluded or delayed contents
Policy / immutable	Root, threshold_hex, leaf masks, or fixed mask	Evaluation labels; mutable per-cycle oracle advice
Observation / controller	Means of charged sources; acquired mask	Unrequested means, cycle identity, cache outputs
Executor / trusted replay	Private traces, price book, cap, request log	Forwarding hidden fields into next_action
Terminal predictor / after STOP	Frozen k, terminal mask, permitted feature coordinates	Choosing actions from other masks' outputs
Auditor / post-hoc	Labels and all frozen predictions for exact scoring	Changing fitted gates or dev choices
Stratum label / post-hoc	Fit-seen truth tuple and root-range flags	Deployment features or new selection inputs

Table 1: Actual restricted replay types; proposed physical or shared-computation extensions are not silently included.

2 S2. Feasibility and normalized quote risk

Mask integers are bit encodings, not ordered amounts of information: 1=PS1, 2=FS1, 4=TS1, 3=PS1+FS1, 5=PS1+TS1, 6=FS1+TS1, and 7=all three. The cost C(m) sums the prices of included sources once. The feasible set is formed before minimizing any loss. Empty mask 0 is retained with its previously defined fallback prediction. With equal prices and cap 2, mask 7 is infeasible even if it has zero prediction error.

For a cycle with e incorrect heads, L=e/4. The scalar objective is u=e/4+λ C(m)/Σc. Under equal quotes, Σc=3; λ=1/20 assigns one extra purchased source a penalty 1/60. The hard cap and lambda serve different purposes. At lambda 0 a two-source action may be chosen despite higher expense, but a three-source action still cannot exceed cap 2. Raising lambda cannot legalize an infeasible action either.

The other inherited price book is (6000, 600, 60), with its own declared caps. These values proxy raw sample counts, not currency or seconds. Scaling both a price book and its cap by a positive constant preserves feasibility and normalized quote loss; scaling the cap alone does not. Changing between

raw-count and equal-source books changes relative prices, so settings must remain separate. Common caps permit different realized average expenditure because a gate may stop on different leaves across cohorts.

$$C(m) = \sum_{j \in m} c_j, \quad \mathcal{F}_B = \{m : C(m) \le B\} \tag{1}$$

$$u_i(m) = e_i \frac{m}{4} + \lambda \frac{C(m)}{\sum_{j=1}^{3} c_j} \tag{2}$$

Mask	Sources	C(m)	Feasible
0	empty fallback	0	yes
1	PS1	1	yes
2	FS1	1	yes
3	PS1+FS1	2	yes
4	TS1	1	yes
5	PS1+TS1	2	yes
6	FS1+TS1	2	yes
7	PS1+FS1+TS1	3	no

Table 2: Equal-source book at B=2. The source and quote contract remains fixed before evaluation.

3 S3. Complete-cycle features and the physical clock

Each local record summarizes an already completed 60-second cycle. PS1 contains 6000 samples, FS1 contains 600, and TS1 contains 60. The policy sees a source mean; the cached terminal classifier uses the declared mean, population standard deviation, and slope features of purchased sources. A mean derived from the completed interval is not an observation available at the beginning of that same interval.

A valid retrospective request means revealing previously recorded data through the permitted interface. It does not mean the equipment could start collecting an unavailable historical source after seeing another completed-cycle mean. A future physical system must specify buffering, overlapping availability windows, action duration, failures after partial work, and whether additional reading changes the device. Those restrictions can change the action set and terminal feature values, so replay validity cannot be established merely by matching a mask.

A general shared-work graph would additionally distinguish work already paid for from features already disclosed. Source reads and transforms could have separate costs and dependencies. The present restricted source-level experiment buys a fixed feature bundle, has deterministic successful reads, and reconstructs its cache from requested sources. It supplies no measured advantage for hidden cache state or general DAG planning.

Source	Samples / completed cycle	Branch input	Terminal bundle	Missing deployment fact
PS1	6000	Mean if purchased	Mean / population SD / slope	When the complete interval can be accessed
FS1	600	Mean if purchased	Mean / population SD / slope	Availability after another source's reveal
TS1	60	Mean if purchased	Mean / population SD / slope	Acquisition delay, failure and buffering

Table 3: Recorded sample inventory and derived timing questions; no measured physical latency is supplied.

4 S4. Frozen predictions and fair policy matching

The reused prediction bank maps (k, mask, cycle) to four categorical outputs. Historical predictor training cycles 1..1540 and train-only standardization remain fixed. Earlier work generated class-balanced neighbor-vote candidates; the gate experiment uses the preserved k1 and k3 predictions. The current diagnostic does not retrain those models or run their forward computations.

A matched acquisition comparison holds k, training bank, feature preprocessing, masks, cap, price book, lambda, cohort, and scoring code fixed. A stronger predictor is a separate intervention, not a benefit attributable to a more elaborate acquisition policy. A historical fixed-source comparator can still be diagnostically important, but its different original selection access must be labeled rather than merged into the primary development selector.

Terminal lookup must occur after STOP. Merely storing all mask outputs is harmless to an auditor; handing those outputs to next_action would change the policy's information rights. The order below

separates the lawful execution from retrospective all-mask evaluation. No out-of-mask features may be used to construct the chosen terminal prediction.

1. Load immutable policy, price book and cap; initialize observed={}, acquired=0, spent=0.
2. Ask next_action using only policy, observed and acquired.
3. If STOP, freeze terminal mask; otherwise reject repeated or unaffordable requests.
4. Charge the source, reveal its allowed mean, append the exact request, and return to step 2.
5. After STOP only, read prediction_bank[k, terminal_mask, cycle].
6. Auditor compares prediction to the target; separate post-hoc code may inspect other masks.

Algorithm 1: Inspected execution order; cached outputs are not policy observations.

5 S5. Gate fitting, development selection, and access rights

The inherited v17 fitting procedure sorts 102 fit means separately for each root and takes candidates at indices floor((102−1)q), q∈{1/4, 1/2, 3/4}. A threshold is rejected unless both branches contain at least 16 fit records. Feasible leaves include the root and at most one other source, and left/right masks must differ. Each legal root/threshold selects a leaf pair by exact fit objective, then lower total quote, lower left mask, and lower right mask.

Development scores the already fitted gates and every cap-feasible fixed mask using the same k and scalar loss. Ties prefer lower quote, fixed before gate, and lexical canonical policy JSON. best_fixed uses fixed masks only; gate_only uses gates, explicitly falling back to best_fixed if no gate exists; adaptive_enabled includes both families. Candidate inclusion proves a weak development-objective inequality, not test dominance.

The original experiment fitted 390 gate-context records across 90 settings; the subsequent v19 diagnostic and this compact v20 revision fit none. Historical evaluation exposure remains even though the current execution writes policy_locks before opening its evaluation bank. Our primary forced gate is labeled as forced because the adaptive-enabled primary selector actually retains PS1.

1. Fix k, price book, hard cap B and lambda before comparing candidates.
2. Construct quartile thresholds from fit means only; retain branches with at least 16 examples.
3. Enumerate distinct affordable root-containing leaf pairs and minimize fit loss with recorded ties.
4. Freeze candidates. Score all feasible fixed masks and frozen gates on the separate dev interval.
5. Choose each family by dev objective, quote, fixed-before-gate and canonical policy order.
6. Record policy locks. Replay the previously exposed evaluation interval without changing choices.
7. Compute privileged oracle diagnostics only after locking; never pass them back to steps 2–5.

Algorithm 2: Historical fit/dev procedure, described for reproducibility; no new fitting is executed for v20.

6 S6. The exact decomposition, one inequality at a time

For cycle i define a_i as the minimum objective across all feasible masks, p_i the minimum across the current gate's two leaves, g_i the actual selected-leaf objective, and f_i the locked fixed-mask objective. Both leaves and the fixed mask are feasible. Therefore a_i≤f_i, a_i≤p_i, and p_i≤g_i. The fixed mask need not belong to the leaf pair; p_i≤f_i is not guaranteed.

Average each quantity over the same nonempty cohort to obtain A,P,G,F. Define H=F−A, E=P−A, R=G−P. The inequalities establish nonnegativity; adding and subtracting the same A and P establishes G−F=−H+E+R. No statistical model or learned optimality theorem is needed. A zero-size support stratum instead retains null means, so this argument is not applied to a fabricated zero-denominator average.

H is headroom against a clairvoyant feasible-mask chooser, not a promised information gain. E bundles mandatory-root acquisition, the leaf set, and their price implications under a fixed predictor. R is loss relative to a label-aware choice among those leaves. Hidden outcome uncertainty, restricted observations, imperfect fitting, and deterministic ties can all contribute. They are accounting gaps, not individually identified causal effects.

For the primary common denominator 25260, the numerator differences are 3360, 2190, 22: −3360+2190+22=−1148. The secondary uses 1684 head decisions because lambda 0 removes quote from its objective: −299+164+72=−63. Mixing these scales would make the decomposition uninterpretable.

$$H = F - A \geq 0, \quad E = P - A \geq 0, \quad R = G - P \geq 0 \tag{3}$$

$$G - F = -(F - A) + (P - A) + (G - P) = -H + E + R \tag{4}$$

Case	F / A / P / G	H	E	R	G−F
Primary	13561 /10201 /12391 /12413, each /25260	56/421	73/842	11/12630	−287/6315
Secondary	924 /625 /789 /861, each /1684	299/1684	41/421	18/421	−63/1684

Table 4: Exact existing cohort arithmetic. G always denotes the actual gate; E always denotes the candidate-constraint gap.

7 S7. Why pair-oracle equality blocks a routing improvement

Let t_i be the TS1-only objective on the primary cohort. TS1 is leaf mask 4; the other leaf is mask 5. Hence p_i=min(t_i,u_i(5))≤t_i for every cycle. The measured equality P=mean(t_i)=12391/25260 implies Σ(t_i−p_i)=0. Every summand is nonnegative, so each must be zero. Any deterministic or randomized chooser between these leaves has expected cycle loss at least p_i=t_i. This proves the stated finite-cohort boundary without assuming that a learned threshold is optimal.

The equality argument requires common positive evaluation weights and identical per-cycle predictor, feasible masks, prices, cap, lambda, and labels. It does not compare unequal objectives or averages from different cohorts. It also does not say which rule wins on future data. If new leaves, a new predictor, altered costs, or a new target loss are introduced, the relevant lower bound must be recomputed and separately evaluated.

A more specific consequence follows from the primary units. Write e5 and e4 for integer head-error counts. Then u_i(5)−u_i(4)=(15(e5−e4)+1)/60, which cannot equal zero. Since p_i=t_i, this difference is strictly positive on every evaluated cycle. The actual gate takes mask 5 on 22 left-branch cycles. Its total excess 22/25260 equals the minimum possible positive sum 22/(60×421), forcing e5=e4 on each chosen left cycle. Thus those 22 extra source purchases account for the entire scalar gap to TS1; equal head-error counts do not imply identical prediction tuples.

This is a deduction from already stored integer counts and exact equality, not a newly executed policy search. The stronger fixed TS1 control has historical selection access and remains explicitly labeled. Its usefulness is to rule out one explanation of the attractive PS1 contrast, rather than to retroactively declare a newly selected primary winner.

$$p_i = \min(t_i, u_i(5)) \le t_i, \quad \sum_{i=1}^{n}(t_i - p_i) = 0 \Rightarrow p_i = t_i \tag{5}$$

$$u_i(5) - u_i(4) = \frac{15(e_i(5) - e_i(4)) + 1}{60} \tag{6}$$

$$G - P = \frac{22}{25260} = \frac{22}{60 \times 421} \tag{7}$$

8 S8. Repeated diagnostics and the missing independent denominator

The dataset contribution is one existing rig and 625 reused cycles:102 fit, 102 dev, 421 exposed evaluation. There are 390 fitted gate-context records but only 36 distinct (k, policy) combinations. Applying every record to all three cohorts produces 1170 rows; four support strata per row produce 4680 subgroup records. Multiplying settings or strata does not create new condition episodes, independent devices, or untouched test units.

Among 390 evaluation gate contexts, the pair oracle cannot strictly beat locked fixed in 147; the actual gate improves 117, ties 14, and worsens 259. There are 126 settings with pair potential unrealized by the gate. Within the 50 gate_only selections, the respective diagnostic counts are 32 unable to beat fixed, 14 actual improvements, 4 ties, 32 regressions, and 4 unrealized-potential cases. The 40 no-gate contexts use fixed fallback and do not enter that 50 denominator.

Among six adaptive-enabled selections that are gates, all six have pair potential; four actual objectives improve and two regress, with two unrealized-potential settings. The other 84 settings retain equal fixed selection. These counts are descriptive across repeated contexts. Their differing k, costs, caps, and lambda values prohibit treating the settings as an independent binomial experiment or averaging scalar objectives indiscriminately.

The alternative verifier imports no author analysis functions and checks 1170 rows, 4680 strata, ten traces, and two altered decomposition fixtures. That is a separate implementation by the same analysis author, followed by a separate root aggregate review; neither is external scientific replication. The execution accounts separately for raw-mean recomputation, cached-loss arithmetic, prediction reuse, and zero new policy fitting.

Quantity	Count	Unit and interpretation
Existing rigs / cycles	1 /625	Not newly collected independent evidence
Gate contexts / distinct k-policy pairs	390 /36	Repeated settings, not390 learned policies in v19
Gate-cohort / subgroup records	1170 /4680	Dependent arithmetic views; empty strata stay null
Gate-cycle diagnostic evaluations	243750	390×625 reused cycle evaluations
New predictor fits / forwards	0 /0	Frozen existing bank
New policy fits / selections	0 /0	Historical v17 choices reused
New independent units / API calls	0 /0	No population or model-run denominator added

Table 5: New arithmetic is separated from reused evidence and new empirical units.

9 S9. Five recorded primary cycles

The primary threshold is exactly 0x1.1b9228afdadcfp+5, printed as 35.44636666666667. Root source TS1 is bought first. A mean≤threshold selects mask 5; otherwise mask 4. The table is transcribed from the immutable v19 traced_cases.json, in its stored explanatory order. F/G/P/A always means locked fixed, actual gate, pair oracle, all-mask oracle.

Cycle 2184 has target(100, 90, 0, 90) and the same displayed prediction(100, 100, 0, 90) for all four actions. The gate reads an unnecessary second source:1/4+2/60=17/60 versus 1/4+1/60=4/15. Cycle 2185 repeats this mechanism. At cycle 1812 the gate and pair oracle make three head errors, while FS1-only all-mask hindsight makes two. That is a leaf-set exclusion witness, not proof that the observable root can select FS1 correctly.

These five cases were selected by predeclared diagnostic categories after outcomes, including positive regret and pair gap. They are not a random sample. Decimal means are displayed faithfully, while hexadecimal means, exact targets and all action predictions remain available in the trace source and the preserved long explainer.

Cycle	TS1 mean; branch→mask	Target tuple	Head errors F/G/P/A	Quotes F/G/P/A	Objectives F/G/P/A
1785	36.2273; right → 4	(100,90,0,130)	2 / 2 / 2 / 2	1 / 1 / 1 / 1	31/60 | 31/60 | 31/60 | 31/60
2184	35.442933333333336; left → 5	(100,90,0,90)	1 / 1 / 1 / 1	1 / 2 / 1 / 1	4/15 | 17/60 | 4/15 | 4/15
1786	36.21248333333334; right → 4	(100,90,0,130)	2 / 2 / 2 / 2	1 / 1 / 1 / 1	31/60 | 31/60 | 31/60 | 31/60
2185	35.43333333333333; left → 5	(100,90,0,90)	1 / 1 / 1 / 1	1 / 2 / 1 / 1	4/15 | 17/60 | 4/15 | 4/15
1812	36.19121666666667; right → 4	(100,73,2,115)	3 / 3 / 3 / 2	1 / 1 / 1 / 1	23/30 | 23/30 | 23/30 | 31/60

Table 6: Existing primary traces; F/G/P/A ordering is identical in the last three columns.

10 S10. Five recorded secondary cycles

The secondary threshold is exactly 0x1.4211a64531f1ap+7, printed as 161.03447166666666. Its PS1 root selects mask 1 on the left and mask 3 on the right. Lambda 0 means all objectives are head-error fractions even though source quotes remain physically relevant proxy accounting. The same immutable illustration rule provides five stored cases.

At cycle 1788 the target is(100, 100, 0, 130). Fixed mask 3 predicts it exactly; the gate stops at mask 1 and predicts(100, 100, 0, 90), causing one head and one vector error while saving one quote. Pair hindsight chooses mask 3, exposing a1/4 routing gap but not establishing learnability from PS1 alone. Cycle 1789 has the same kind of tradeoff. At cycle 1787 even the pair retains two errors while all-mask FS1 hindsight has one.

Across all 421 evaluation cycles the gate has 861 rather than 924 component errors and 518 rather than 842 quotes, but 391 rather than 381 vector errors. The five cases explain mechanisms; they cannot replace those complete-cohort counts. An error objective that rewards partial correctness and an objective requiring every head correct answer different decision questions [19].

Cycle	PS1 mean; branch→mask	Target tuple	Head errors F/ G/P/A	Quotes F/G/P/A	Objectives F/G/ P/A
1785	161.06345166666665; right → 3	(100,90,0,130)	2 / 2 / 2 / 2	2 / 2 / 1 / 1	1/2 | 1/2 | 1/2 | 1/2
1788	161.01096166666667; left → 1	(100,100,0,130)	0 / 1 / 0 / 0	2 / 1 / 2 / 2	0 | 1/4 | 0 | 0
1786	161.05212166666666; right → 3	(100,90,0,130)	2 / 2 / 2 / 2	2 / 2 / 1 / 1	1/2 | 1/2 | 1/2 | 1/2
1789	161.01089166666665; left → 1	(100,100,0,130)	0 / 1 / 0 / 0	2 / 1 / 2 / 2	0 | 1/4 | 0 | 0
1787	161.05556666666666; right → 3	(100,90,0,130)	2 / 2 / 2 / 1	2 / 2 / 1 / 1	1/2 | 1/2 | 1/2 | 1/4

Table 7: Existing secondary traces; quote savings survive lambda0 accounting but are not measured deployment savings.

11 S11. Root ranges, label support, and cohort movement

All 102 fit labels share one target tuple. It occurs in 2/102 dev cycles and 30/421 evaluation cycles. This restricts what the fit stage has seen about condition-dependent utility; it does not prove all observation-based learning is impossible. Predictor mistakes can vary within one target condition. Whether such relations survive new conditions is an untested transport question.

For primary g129, branch counts shift 26/76 in fit to 0/102 in dev and 22/399 in evaluation. Its root means leave the fit minimum–maximum range in 100 dev and 392 evaluation cycles. Secondary g301 shifts 51/51→85/17→324/97; its corresponding outside-range counts are 99 and 382. Cohort-dependent realized quote totals follow these branch frequencies despite an unchanged per-cycle cap.

Range membership is a one-dimensional descriptive flag. A point inside the range may lie in a low-density gap; a point outside may be predicted accurately. Joint feature support, device identity, concept drift, uncertainty calibration, and causality require more than a min/max test. Truth-tuple seen/unseen strata use audited labels and cannot become deployment inputs.

The primary 30 fit-seen evaluation cycles have zero all-mask headroom and actual-minus-fixed 19/1800 due to extra quote. The 391 fit-unseen cycles have difference−389/7820. Both are post-hoc decompositions on an exposed cohort; neither authorizes a new label-aware selector. Prospective training should diversify conditions before reserving an untouched device or regime.

Gate / cohort	n	L / R	Root outside fit range	Quote fixed / gate
g129 fit	102	26 /76	0	102 /128
g129 dev	102	0 /102	100	102 /102
g129 evaluation	421	22 /399	392	421 /443
g301 fit	102	51 /51	0	204 /153
g301 dev	102	85 /17	99	204 /119
g301 evaluation	421	324 /97	382	842 /518

Table 8: Stored cohort movement. L/R are gate branch counts; no row is a new independent rig.

12 S12. What the library comparators would actually test

A method name is not an implementation. A native reproduction preserves its original architecture, objective, data and evaluation protocol; a common-contract adapter changes some of them and must disclose the changes. A handcrafted reference gate claims neither. The table identifies which missing capability each comparison could test and the work not performed here.

Fixed predictor matching is especially important for BRiG-AFA: comparing its learned future-risk policy against a different predictor would entangle acquisition and classification. Generative surrogates and EDDI require a model of unobserved outcomes, not just a confidence scalar. Learned mutual-information selection supplies a meaningful greedy competitor before nonmyopic complexity is credited. Hierarchical costly features require shared structure and acquisition costs that our three flat sources do not instantiate.

The inherited reference objects retain their exact number, text, URL and claim_scope. Some recorded scopes cover only official abstracts, while selected others describe reviewed sections. This compact reorganization adds no fresh primary-source check or native benchmark result. Structured validation [14–16], offline acquisition assumptions [11], selective risk/coverage [6, 17, 18], and adaptive-submodular conditions [7] remain limits rather than borrowed guarantees.

Reference family	Strength worth testing	Required matching / unimplemented part
Joint AFA/classification [8,21]	Learned acquire-or-classify behavior	Separate predictor and policy training effects
Hierarchical costly features [20]	Structured source dependencies and costs	Actual hierarchy, shared-cost semantics, native agent
GSM [9] / EDDI [12]	Conditional missing-feature models	Fit and validate forecasts without evaluation leakage
Learned information gain [10]	Strong amortized greedy selection	Same information, predictor, loss and total training budget
Learning-To-Measure [2]	Predictor/acquisition coupling	Native predictor/meta-training and evaluation rights
BRiG-AFA [3]	Remaining-budget risk-to-go	Frozen predictor; original budget and fitted future-risk model
O-MPAC [4] / NM-PPG [5]	Planning or trajectory optimization	Native teacher/scorer or gradient machinery and costs
Survey [13]	Taxonomy for organizing methods	Taxonomy alone does not validate any implementation

Table 9: Methodological comparators, all unexecuted natively in this local study.

13 S13. Negative controls, latency accounting, and falsifiable next evidence

Existing trusted-interface poison tests hold acquired values fixed while changing hidden values; unchanged actions support the declared observation boundary. Existing checks also reject altered decomposition fixtures and rehash original prediction, policy and raw-source inputs. These tests target finite invariants, not physical acquisition, malicious-process isolation, calibrated uncertainty, or scientific validity.

Realized quotes count requested sources under a declared book. They do not include full-record collection, training data acquisition, policy fitting, cache construction, tuning, hardware startup, failure recovery, network transport, or wall-clock delay. A comparison can obey the same hard cap and still spend more per cycle. Future methods must report their training and inference budgets as well as terminal acquisition quotes.

A falsifiable prospective study would reserve a new device or time regime before fitting, diversify the policy-fit target conditions, and predeclare component and vector endpoints. Freeze source availability, failure rules, exact cap, prices, lambda, predictor controls and tuning allocations. Compare strong fixed TS1 where appropriate, exhaustive feasible fixed masks, the small gate, and one explicitly native or adapted literature method. Evaluate policy behavior only on information available at its real decision time.

A gain claim fails if the locked selector does not improve the prespecified endpoint against the strong matched control, violates acquisition timing, or hides a prespecified vector regression or failed request. Report empty coverage and missing strata, rather than assigning them zero error. This is a proposed admission study; no new cohort, model, acquisition, statistical coverage, or professor endorsement is claimed.

Check	Current status	What a new study must supply
No hidden-value action dependence	Inspected trusted-code fixtures	Actual deployment permission and timing tests
Exact F/A/P/G identity	Saved arithmetic and altered-fixture checks	Same contracts on a newly reserved cohort
Strong fixed-source control	Historical TS1 retained with selection scope	Locked common selection access and budgets
Component versus vector conflict	Observed secondary -63 /$+10$	Predetermined endpoint and tradeoff acceptance
Source cost and latency	Quotes only; real latency unmeasured	Availability, failures, wall-clock/ energy/currency records
Independent generalization	Not established	New device/regime, provenance and untouched evaluation

Table 10: Prospective requirements are unexecuted; saved checks are labeled separately.

14 Complete preserved v19 scientific record

The following complete record preserves the earlier scientific sections, negative results, equations, tables, algorithms, figures and appendices. Its development-stage statements retain their original

scope. The current compact conclusions and detailed derivations precede this archived material; retaining an earlier result does not turn it into an independent replication.

ProbeFlow: Observation-Dependent Acquisition Exposes Selection, Cost and Diagnosis Tradeoffs: Frozen predictors, chronological policy fitting and inspectable acquisition traces

Neutral working paper v19, 2026-10-19. New finite saved-evidence diagnostics; no new independent data, real model/API calls, native-method trial or physical acquisition. Previous v17 evidence and v18 presentation are retained.

Observation-dependent acquisition needs a stronger control than a favorable fixed-mask comparison. ProbeFlow v17 fits 390 small acquisition gates on 102 chronological policy-fit cycles, selects among gates and all feasible fixed subsets on 102 policy-dev cycles, and replays frozen terminal predictions on 421 previously exposed evaluation cycles. The prelocked grid has 90 predictor, price, cap and cost-weight settings. At the primary equal cap 2, weight 0.05 and k1, the adaptive-enabled selector retains fixed PS1: 876/1684 component errors and 391/421 vector errors. A forced TS1-first gate reaches 798 component errors but only matches the existing TS1/k1 control while spending 22 extra quote units. Six settings select a k3 gate; four lower component error, two raise it, and the remaining 84 tie fixed selection. Component gains can accompany ten additional vector errors, while a three-count dev gain can reverse into 62 additional evaluation component errors. All 102 policy-fit targets share one joint condition, sharply limiting learned condition-dependent utility. We preserve this support failure, exact request traces and nonzero policy-learning counts. These are exposed offline policy executions, not an independent final, physical cost saving or native AFA superiority claim. A new exact oracle-separated diagnostic shows that the primary gate's two frozen leaves cannot strictly beat the historical TS1 objective on the exposed cohort, while a secondary component benefit retains vector regression.

- A bounded, executed four-objective selection audit on one frozen predictor and prediction bank, separating new decision analysis from new predictions and independent trials.
- A conditional training-tuple output-support ceiling, with a separate empty-fallback check and explicit budget-achievability and forced-prediction conditions.
- Matched loss/cost tables, projected Pareto diagnostics and same-loss hindsight fixed-subset regret that retain unfavorable transfers and head-level tradeoffs.
- A reproducible continuation of the v11/v12 support-first study with eight boundary fixtures, consumed code/input snapshots and a long-form Korean explanation; no industrial, adaptive or native-method superiority claim.
- A real matched compositional neighbor baseline separates output-class expressibility, selection-locked utility and exposed-grid diagnostics, with exact 5,000-record baseline reproduction and a retained negative selection result.
- A matched fixed-policy benchmark distinguishes globally averaged predictor selection from budget-conditioned joint selection, retaining unchanged source masks, vector failures, stronger fixed-source controls, cost-sensitive counterexamples and chronological selection instability.
- A real observed-value acquisition gate benchmark separates policy fitting from frozen predictor reuse, compares all feasible fixed subsets, preserves strong TS1 controls and reports metric conflicts, expenditure drift and policy-fit support failure.
- A new post-hoc diagnostic holds all390 fitted gate-context records, the frozen predictions, prices, hard cap, k and lambda fixed. For each cohort, A is the per-cycle all-feasible-mask oracle mean, P the current two-leaf oracle mean, G the actual gate mean and F the locked fixed-policy mean. The exact identity G-F=-(F-A)+(P-A)+(G-P) splits the difference into negative headroom, the root/leaf/cost-constraint representation gap and privileged routing regret. The last two are not pure learning errors or causal effects: the oracle observes labels and unseen predictions unavailable to an executable policy.

15 Preserved v19: Introduction

Active feature acquisition decides which information to request before making a prediction. More elaborate policies can reason over future observations, remaining budgets, and stop decisions. Yet a policy comparison can be uninformative when its fixed predictor cannot exploit even the complete permitted input. ProbeFlow studies that dependency in a public sensor replay: what should be checked before spending additional effort on acquisition search?

The practical waste is specific. A longer search tree requires future-observation estimates, additional inference, parameter choices, and a larger audit surface. If every resulting terminal diagnosis abstains, or if its error is unchanged across source masks, a lower planning objective cannot establish useful

diagnosis. We therefore diagnose data support, predictive utility, and terminal coverage before escalating acquisition complexity. This ordering is a proposed research discipline illustrated by the present sequence; we have not randomized research teams or measured its general effect on research productivity.

The public Hydraulic dataset contains recorded 60-second cycles and four categorical condition targets [1]. Our local assets contain PS1, FS1, and TS1 only. We do not evaluate the full physical sensor bank. The current task is retrospective classification of completed recorded cycles, not online equipment control or causal fault intervention. Reading an offline file does not demonstrate that the same sensor request would have the modeled cost or measurement effect in a deployed system.

All numerical results below are local computations on these assets. Earlier finite examples, the v11 sensor replay, the v12 predictor comparison, and author-reported literature results remain distinct evidence classes. The 421 evaluation cycles were already inspected in v9 and v11; locking a new program before its current evaluation pass supplies a reproducible procedural boundary, not renewed blindness. We use no independent-sample confidence intervals or population significance tests.

V14 asks a second, decision-level question: after a predictor is frozen, can changing its selection loss repair the diagnosis, and which component pays for the change? We freeze four objective rules and analyze their decisions on the already preserved prediction bank. This is intentionally post-hoc development research: earlier aggregate outcomes were known when this question was formulated. A new plan and execution snapshot make the continuation inspectable; they do not make its evaluation independent. The distinction between a better scalar objective and a representable complete diagnosis becomes the central limit.

Figure 1 locates the observed failures in this sequence. The support check remains qualified even after marginal class presence is restored. The subsequent bounded predictor contrast supplies a reason to run small, matched acquisition controls, while its unfavorable results limit further claims. A richer information-state contract remains a hypothesis for a later, properly supported experiment.

The v14 ceiling assumes that predictions copy training joint tuples. An independently decoded model can violate this assumption without adding a sensor. V15 makes that possibility executable by balancing neighbor votes separately for each target head. The decisive outcome has two parts: some unselected configurations produce correct unseen tuples, but the one admitted by selection remains the old k=1 rule. Keeping those findings separate prevents a retrospective counterexample from becoming an unsupported performance victory.

V16 asks whether that v15 outcome is specific to its mask-averaged selection criterion. Budget-conditioned joint model/mask selection is a stronger fixed control, yet changing the predictor must not be credited to acquisition planning. The resulting local component gains coexist with unchanged source masks and vector failures, a stronger fixed-source comparator, and unstable choices across dependent chronological views.

V17 moves from fixed masks to a bounded observation-dependent policy. Its primary selector does not choose adaptation; a forced gate's apparent component gain vanishes against the preserved TS1 control. Sensitivity outcomes expose complete-diagnosis conflicts and a homogeneous policy-fit interval. The contribution is an executed, auditable acquisition comparison with these unfavorable boundaries, not a general adaptive-policy victory.

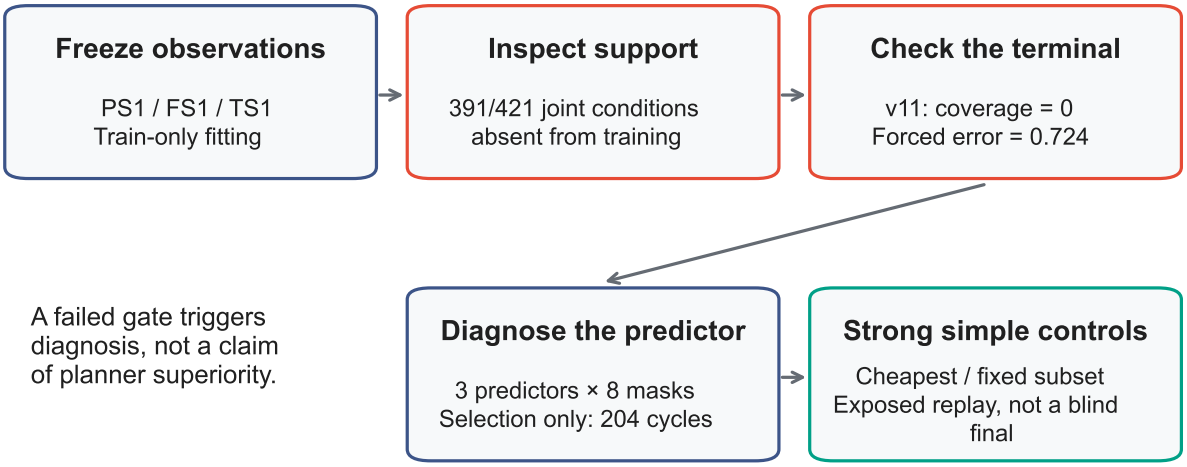


Figure 1: Diagnostic ordering and the observed v11-to-v12 detour. Zero selective coverage leads to a bounded predictor audit before additional planner search. Arrows describe the study logic, not a proven causal benefit or universally sufficient checklist.

16 Preserved v19: Related Work

Our closest comparison is active feature acquisition (AFA): a predictor and an information policy are evaluated together under acquisition constraints. A recent survey organizes this literature using partially observed decision problems and model-based, model-free, and hybrid approaches [13]. We use that taxonomy to locate the study, while separating conceptual similarity from an executed comparison. External benchmark scores are not inserted into our local result tables.

16.1 Preserved v19: Learned policies, surrogates, and strong greedy controls

Joint acquisition and classification can learn a cost-sensitive stop-or-acquire agent alongside a variable-size representation [8]. EDDI uses a partial variational model and expected information gain [12]; generative-surrogate AFA supplies models of missing features to support decisions [9]. These approaches clarify why a current confidence value is not a distribution over unobserved outcomes. ProbeFlow does not train these generative components or test their conditional forecasts.

Amortized conditional-mutual-information selection provides a learned greedy alternative [10]. Its relevance here is comparator strength: a small or greedy policy deserves careful construction before a longer horizon is credited. Our cheapest-first, pressure-first, and exhaustive eight-subset development controls are simpler still. They are useful within this small action space but cannot substitute for the native objectives, architectures, or training resources of these methods.

16.2 Preserved v19: Recent nonmyopic and grouped-source methods

Learning-To-Measure combines an in-context predictor with acquisition and includes greedy and lookahead variants [2]. BRiG-AFA learns remaining-budget risk-to-go for a fixed predictor [3]. O-MPAC learns source-acquisition decisions from offline planning [4]. NM-PPG optimizes nonmyopic trajectories with pathwise policy gradients [5]. These primary texts establish that lookahead, fixed-predictor planning, and grouped acquisition are existing design choices. None alone is a novelty claim here.

We distinguish three possible comparisons. A native reproduction preserves the original method and evaluation contract. A common-contract adapter may change actions, prices, terminal loss, or predictor and must disclose those changes. A handcrafted reference policy makes no claim to implement a named method. Our v11 empirical planner belongs to the third category. Native reproduction counts remain zero; a future comparison must separately budget training, teachers, forecasts, and tuning.

Reference	Relevant design feature	Missing native reproduction component
Learning-To-Measure [2]	Predictor/acquisition coupling; greedy and lookahead	Native meta-predictor, training, retrospective-support assumptions
BRiG-AFA [3]	Fixed predictor and learned future risk	Backward fitted networks and original acquisition objective
O-MPAC [4]	Training-record teacher and receding source planning	Original teacher/scorer training and native data protocol
NM-PPG [5]	Trajectory objective and nonmyopic optimization	Pathwise optimization, native transition and cost model

Table 11: Method comparisons are conceptual and contract-level. No source-reported benchmark number is placed in our locally measured performance tables.

16.3 Preserved v19: Selective decisions, offline evaluation, and structural assumptions

Selective classification makes coverage part of the outcome [18]. SelectiveNet jointly learns prediction and rejection [17], whereas our v11 fixed predictor uses a development-selected threshold. SGR supplies a risk-control construction under its sampling and selection assumptions [6]; the present threshold is not such a certificate. Empty coverage leaves selective error undefined.

Offline AFA evaluation can depend on how data were acquired and on assumptions connecting observed and deployed acquisition [11]. Adaptive-submodularity guarantees likewise require a specified distribution and objective satisfying the theorem conditions [7]. A complete-record replay with a mask lookup is not automatically a valid evaluation of arbitrary adaptive histories, and our fixed classifier is not covered by a generic guarantee of monotone empirical error.

16.4 Preserved v19: Loss-specific decisions and the comparison gap

Loss-specific decisions and Hamming/subset-zero-one tradeoffs are established topics [19]. We examine a finite four-head multiclass acquisition system rather than reproducing binary-label theory. Our added question is operational: which selection-locked subset changes under declared terminal losses, and can that change escape this predictor's actual output support? No new general loss theory or AFA optimizer is claimed.

17 Preserved v19: Problem and Diagnostic Method

A latent record consists of complete sensor traces and the condition label. The controller initially receives the declared source catalogue, legal action set, quote map, budget, and any explicitly public history. It may receive a statistic only after the corresponding successful action. The true evaluation label, unrequested numerical features, candidate future outcomes from the evaluation record, and private cache intermediates are excluded from controller inputs.

In the general contract, state must distinguish publicly disclosed features from work already completed by the executor. A cached raw record need not mean that every derived statistic has been disclosed. Reusing an already-read source can change a marginal quote without granting free access to every downstream transform. This is a contract requirement, not an established empirical advantage of a new state representation.

The present three-channel experiment is a restricted case: requesting one channel discloses all three predefined statistics together, requests succeed deterministically, and each source has a fixed quote. The cache is reconstructible from the requested-channel history. In this case the observed mask, disclosed values, remaining budget, and catalogue suffice for the implemented policy comparison; we do not claim that an unobserved cache variable provides additional predictive power. Partial windows, overlapping sources, failed reads after partial work, or separate transform requests require a subsequent experiment.

A planner additionally needs a predictive distribution for future observations. Current confidence is not that distribution. The v11 empirical planner used complete training prototypes to forecast possible source outcomes; it never used unrequested evaluation values as its forecast target. Those empirical forecasts may be wrong under joint-condition shift. The v12 study deliberately steps back to fixed mask baselines so that a predictor failure cannot be mistaken for evidence favoring or disfavoring a sophisticated transition model.

$$J(p) = E[L + \alpha C] \tag{8}$$

$$C(A) = \sum_{g \in G(A)} c_g \tag{9}$$

17.1 Preserved v19: Unit-aware source quotes and hard budgets

For a selected source, mean, population standard deviation, and slope share one source-read quote. The raw-sample price system assigns PS1=6000, FS1=600, and TS1=60. A separate equal-source system assigns one unit to each channel. These units are never summed across price systems. A preflight failure consumes no quote and discloses nothing. Repeated statistics of the same acquired source are not charged as independent raw reads.

The quote is a logical replay proxy, not a measured physical I/O, energy, monetary, or end-to-end latency cost. In fact, the offline evaluator loads the complete feature bank to support all mask projections; the quote records the information rights assigned to a policy, not which files the audit host physically avoided reading. Training, feature-bank construction, and inference wall time are reported separately. Substituting a source-count quote for a raw-sample quote changes the decision problem, even if the predictor remains fixed.

Our scalar objective is component error plus 0.05 times mean quote normalized by the sum of the three source prices. The normalization and coefficient were fixed before the v12 run. They are not fitted after observing a favorable ranking. Hard budget feasibility is enforced separately from the soft penalty; a low expected penalty cannot license an over-budget action.

17.2 Preserved v19: Diagnostic admission and selective denominators

The admission sequence asks different questions rather than collapsing them into one pass flag. Are the source bytes and label definitions pinned? Are the evaluation marginal labels and joint conditions represented in training? Does a fixed predictor improve on no evidence under complete and partial masks? Does the terminal rule make accepted predictions? Only then is a policy-specific comparison interpretable. An unmet support condition is disclosed, not silently repaired by excluding hard cycles.

We measure mean four-head component error, complete-vector error, terminal coverage, and scenario-specific quote. Component error counts four decisions per cycle; vector error counts a cycle incorrect if any head is wrong. These denominators are descriptive and correlated. Coverage is accepted cycles divided by all declared cycles. Selective risk divides accepted errors by accepted decisions and is null when none are accepted. The v12 predictor audit makes no selective-risk claim and does not replace the v11 terminal result.

$$R = \frac{e_1 + e_2 + e_3 + e_4}{4n} \tag{10}$$

$$R_{\text{vec}} = \left(\frac{1}{n}\right) \sum_{i=1}^{n} \mathbf{1}(\hat{y}_i \neq y_i) \tag{11}$$

$$\text{coverage} = \frac{n_{\text{accepted}}}{n} \tag{12}$$

17.3 Preserved v19: Loss contracts and a closed-output support ceiling

Let e_ih indicate an error on head h of cycle i. Component loss averages the four head decisions; vector loss charges a cycle once if any head is wrong. Worst-head loss is the maximum of the four aggregate head error rates, not the mean of a per-cycle maximum: that latter quantity would equal vector loss. Pump loss counts errors on the third target only. All four outputs are proportions but their decision units and priorities differ. Differences between rules are meaningful within the same loss, cycles and price system; subtracting vector loss from component loss is not an improvement measure. The elementary bounds component<=vector<=min(1,4*component) leave substantial room for mismatched rankings. Our boundary fixture constructs an explicit ranking reversal; it is an authored illustration rather than a hydraulic result.

A sharper conditional restriction follows from the output class. Let T be the set of complete training label tuples. For every nonempty observed mask, the fixed 1NN returns all four labels of one training cycle, so its output lies in T. An empty mask is different: its per-head modal fallback can combine labels into a tuple absent from T. The closed-support statement therefore requires separately checking that fallback. In this cohort it is (3,100,0,90), and it belongs to T. A synthetic counterexample with training tuples (1,0,0,0), (0,1,0,0) and (0,0,1,0) has modal tuple (0,0,0,0) outside support; our verifier rejects applying the training-support bound in that case.

Conditional proposition. For any forced complete-vector decision rule whose every legal terminal output belongs to T, the number of exact successes on a finite cohort cannot exceed the number of true tuples belonging to T. Proof: when a true tuple is outside T, equality with any allowed output in T is impossible. Sum that zero-success fact over unsupported cycles; supported cycles contribute at most one each. The argument needs no sampling independence. It is conditional on the

output restriction, not on an assumption that unseen combinations are intrinsically undiagnosable. Head-wise independent classifiers, a changed representation/decoder, or an empty fallback outside T need not satisfy the premise. A stop or abstention is not an exact successful vector. The bound is an upper bound, not a promise that a legal budget can attain it. The inherited support-bound audit did not execute an adaptive rollout; the set argument describes a hypothetical rule retaining this restricted output class, not its measured policy performance. V17 below separately executes observation-dependent offline replay.

$$R_C = \left(\frac{1}{4n}\right) \sum_{i=1}^{n} \sum_{h=1}^{4} e_{ih} \tag{13}$$

$$R_V = \left(\frac{1}{n}\right) \sum_{i=1}^{n} \mathbf{1}\left(\sum_{h=1}^{4} e_{ih} > 0\right) \tag{14}$$

$$R_W = \max_{h=1,\ldots,4} \left(\left(\frac{1}{n}\right) \sum_{i=1}^{n} e_{ih} \right) \tag{15}$$

$$R_P = \left(\frac{1}{n}\right) \sum_{i=1}^{n} e_{i3} \tag{16}$$

$$R_C \leq R_V \leq \min(1, 4R_C) \tag{17}$$

$$\text{exact successes} \leq \sum_{i=1}^{n} \mathbf{1}(y_i \in T) \tag{18}$$

18 Preserved v19: Experimental Setup

The inherited v11/v12 experiments remain unchanged. V13 added source review and manuscript structure without new trials. V14 adds one bounded offline objective/decision analysis of the preserved prediction files. It does not fit a model, call a predictor, collect a physical cycle, train a literature-native policy or acquire an independent final set. The following inherited setup is followed by the explicit v14 continuation.

18.1 Preserved v19: Data, chronology, and support

The raw assets are byte-identical to the earlier registered subset: PS1 contains 6000 values per cycle, FS1 600, and TS1 60; profile supplies the four condition labels. We preserve all 2205 cycles as the data inventory, although only the declared train, selection, and evaluation intervals are used by the comparison. The feature extractor computes the same three statistics per channel as v9/v11. Label fields, the stability flag, and cycle index are not predictive features.

The v9 chronological training prefix ended at cycle 1323 and contained no cooler=100 example; all 421 evaluation cycles had cooler=100. The v11/v12 prefix ends at 1540 and contains cooler counts 732, 732, and 76 for classes 3, 20, and 100. Each evaluation head class is now present in training. This repairs a marginal-label admission failure but does not establish support for the joint diagnostic question.

All 76 training records with cooler=100 have the same four-head condition tuple, (100,100,0,90). The evaluation segment contains that tuple only 30 times. Therefore 391 of 421 evaluation cycles, or 92.87%, have joint conditions absent from training. Across all classes, the training set contains 97 distinct condition tuples, selection 11, and evaluation 38. The 194 contiguous same-condition runs in the full profile are not established independent experimental units or physical assets.

This distinction prevents a misleading remedy: merely showing every head label somewhere in training does not supply examples of every relevant mechanism combination. Our reported errors remain well-defined descriptive errors on the exposed interval, but their scientific interpretation is conditional on this narrow single-rig support. A random cycle split could mix adjacent experimental regimes; we do not introduce one to manufacture an easier final result.

Structured validation should match the intended deployment problem: blocking may reduce dependence but can also change interpolation into extrapolation [16]. Our two 20-cycle gaps are transparent exclusions, not evidence that adjacent retained cycles are independent. The absence of verified asset or physical-session identities prevents us from defining a justified independent-unit count.

Partition	Inclusive cycles	Count	Permitted use
Training	1-1540	1540	Scaler, centroids, nearest-neighbor reference bank
Embargo	1541-1560	20	Excluded
Selection	1561-1764	204	One model choice and predeclared subset selection
Embargo	1765-1784	20	Excluded
Exposed evaluation	1785-2205	421	Fixed comparison and descriptive audit

Table 12: The embargo preserves temporal ordering; it does not prove statistical independence. Current evaluation decoding occurs after the new selection artifact is frozen, but earlier exposure remains.

18.2 Preserved v19: Frozen predictor contrasts and selection rights

We limit the candidate set to three simple predictors. The prior-weighted centroid is exactly the existing v9/v11 rule: training-only feature means and population standard deviations; one standardized centroid per head/class; average squared distance over disclosed dimensions; and a log training class-prior offset. Each head is classified independently. Tests compare this implementation against the original predictor on every mask.

The prior-neutral centroid uses the same fitted means, scales, centroids, and masked distance. For nonempty masks it removes only the log-prior offset. This is a controlled software scoring contrast, not a claim that a balanced prior is correct for an industrial population. The empty-mask fallback is held common across all candidates: the modal training class per head, with numerical label ordering for ties.

The one-nearest-neighbor predictor searches complete training cycles using standardized distance in the disclosed feature subspace and returns the winning training cycle's four labels. Exact distance ties choose the earliest training row; the index is a deterministic tie rule, not an input feature. No k, metric, preprocessing, or mask-specific fit is tuned. This changes local representation, multi-head coupling, and inductive bias together, so its difference from a centroid cannot uniquely identify one representation mechanism.

All eight source subsets, including the empty set, are evaluated with one shared training fit per candidate family. We select exactly one predictor using the equal mean selection component error across the eight masks; ties follow the declared candidate order. No evaluation outcome enters that choice. The selection criterion intentionally differs from choosing the best all-source score, because acquisition systems operate under partial observations.

Finite-sample model selection can itself overfit a development criterion [14,15]. We therefore preserve both rights boundaries and their limits: the same 204 labels support the model choice and the permitted subset choices; the subsequent segment was exposed in earlier work. Freezing the current selection artifact before current evaluation decoding makes the run reproducible, but does not turn the interval into a fresh holdout.

Pin plan, raw hashes, dependency hash, and exact consumed code in a new output directory.
Fit one common scaler and training-only reference parameters on cycles 1-1540.
Evaluate the three declared predictors under all eight masks on the 204 selection cycles.
Select one predictor by mean selection mask error; freeze the choice and any permitted subset masks.
Decode the current evaluation segment; record all 24 fixed model-mask prediction groups.
Reconcile recorded predictions, head errors, vector errors, mask prices, and denominators.
Retain negative comparisons, price-dependent reversals, and the earlier exposure record.

Algorithm 3: One bounded run; no additional candidates are proposed after its evaluation results.

18.3 Preserved v19: Conditional acquisition controls

Before evaluation, the selected predictor must improve its full-source selection component error by at least 0.05 over the common no-evidence baseline, with no invalid prediction. If this usefulness gate fails, no policy search is added. The gate is a development eligibility rule, not a statistical certificate. It prevents a longer planner from being evaluated against a system that cannot use even the complete declared input.

If eligible, four fixed policies share the selected predictor: stop; cheapest-first; pressure-first; and selection-locked subset. The last policy considers all eight feasible masks and chooses the lowest selection error plus normalized quote. Ties prefer lower quote and then mask index. It is a strong small-design baseline, not a new adaptive algorithm. Cheapest-first uses price order with deterministic lexical ties; pressure-first uses PS1, FS1, TS1 and skips unaffordable sources.

Raw-sample budgets are 0, 60, 660, 6000, and 6660; equal-source budgets are 0, 1, 2, and 3. Every chosen mask satisfies the corresponding hard cap. Model selection and subset selection reuse the same

204 development labels, which can overfit their conditions. We report that reuse explicitly rather than introducing an unobserved calibration or final set.

The inherited fixed-subset evaluation reuses the saved predictions of the frozen model-mask groups. It does not invoke a new predictor with test labels and does not generate additional independent outcomes. For that inherited comparison, this reuse is valid because its policies are fixed subsets. It would not justify counterfactual replay of an adaptive policy whose information history or state transition was absent from the trace. V17 below separately supplies observed-value requests and terminal masks for its offline replay under the stated completed-cycle assumptions.

18.4 Preserved v19: V14 objective-audit protocol and information rights

We inherit the previously selected 1NN rather than selecting a new predictor for each objective. The four rules are component, vector, worst-head and pump; all use the same 204 selection labels, eight masks and nine price-budget contracts. Each rule minimizes its own selection loss plus 0.05 times quote normalized by the sum of source prices, subject to the same hard cap. The coefficient is reused as a declared sensitivity contract, not an elicited equivalence between the four utility scales. Exact rational arithmetic fixes ties by lower quote and then mask index. The original component-selected masks are reproduced in all nine contracts.

Before computation, the new directory receives the plan, exact consumed runner, both prediction JSONLs, profile labels, prior frozen selection artifact and prior result summary. The selection function is passed selection aggregate metrics only, and its 36 decisions are saved before current evaluation metrics are computed. The audit infrastructure holds exposed profile bytes; this is a procedural access boundary, not blindness. The new 36 rows equal four objective rules times nine contracts. They differ from the earlier 36 rows, which were four policy baselines times nine contracts. Both reuse the same 421 cycles, and neither is a sample of 36 independent experiments.

We compute a projected Pareto frontier in component error, vector error and quote, separately within each partition, price system and budget. Strict dominance requires no worse values in all three coordinates and a strict gain in at least one; equal points stay tied. This projection does not include pump or worst-head utility. A pump-optimal subset may therefore be dominated in the projection without being inferior for its declared pump objective.

For each target loss we additionally report a hindsight best single fixed subset using exposed evaluation labels. Same-loss penalized regret is the chosen rule's evaluation loss plus normalized quote minus this optimistic minimum. Its nonnegativity follows from the minimum, not from successful generalization. It is not a deployable selection rule, an adaptive per-cycle oracle or independent evaluation. True joint labels also define seen/unseen retrospective strata; policies never receive evaluation stratum membership. Empty stratum rates would be null.

$$m_l = \arg \min_{m:c(m)\le B} \left(R_{l,\mathrm{sel}}(m) + 0.05 \frac{c(m)}{c_{\mathrm{all}}} \right) \tag{19}$$

$$\mathrm{regret}_l(m) = R_{l,\mathrm{eval}}(m) + 0.05 \frac{c(m)}{c_{\mathrm{all}}} - \min_{a:c(a)\le B} \left(R_{l,\mathrm{eval}}(a) + 0.05 \frac{c(a)}{c_{\mathrm{all}}} \right) \tag{20}$$

19 Preserved v19: Results and Analysis

Results follow the diagnostic order rather than presenting a single favorable aggregate. The earlier selective failure, the bounded predictor change, the source-mask geometry, and the price-dependent policy controls answer different questions. Every reported local arm remains in its original denominator.

19.1 Preserved v19: V11: repaired marginal labels, zero terminal coverage

Restoring marginal class presence in v11 did not produce useful selective diagnoses. The selection-tuned common threshold was 1.01, so all 47 policy-budget-price groups had zero coverage and undefined selective error; their terminal abstention loss was 0.3. A separate, explicitly posthoc one-run diagnostic forced the threshold to zero. It was not substituted for the main selective result.

Under that diagnostic and raw budget 6660, empirical greedy and two-step lookahead acquired mean quotes 269.50 and 2419.38, respectively. The complete three-source reference cost 6660. All had forced component error 0.723872 and vector error 1. The longer planner increased the quote without predictive benefit. Planning plus replay wall time was approximately 2.48 seconds for greedy, 23.03 for two-step lookahead, and 0.015 for stop on that single sequential host run; these times are not randomized performance estimates or physical measurement costs.

This failure does not refute nonmyopic acquisition in general. It identifies a concrete dependency of the experiment: if the predictor and its forecast provide no useful terminal decision on the target segment, additional search is not evidence of improved information use. We preserve this unsuccessful stage rather than presenting the later predictor contrast as if it had been the initial selective hypothesis.

19.2 Preserved v19: V12: a full-source gain confined to one head

The selection mean mask errors were 0.555147 for the prior centroid, 0.348499 for the prior-neutral centroid, and 0.339308 for one-nearest-neighbor. The declared rule selected one-nearest-neighbor. Its full-source selection error was 0.305147 against a no-evidence error of 0.555147, so the 0.05 usefulness gate passed with an observed development gain of 0.25.

On the exposed evaluation interval, the prior centroid had component error 0.723872 under every mask. Removing the prior offset reduced full-source error to 0.473872; one-nearest-neighbor had the same full-source error. In both improved full-source cases the cooler head had zero errors, while valve, pump, and accumulator errors remained 273, 246, and 279 out of 421. Thus the full-source 0.25 reduction is exactly one corrected head per cycle, not broad four-component diagnostic recovery.

The full-source nearest-neighbor prediction was the same tuple, (100,100,0,90), on all 421 cycles. Thirty cycles matched it completely and 391 did not, giving vector error 0.928741. The improved component score therefore coexists with a high rate of incorrect complete diagnostic vectors. Reporting only the component mean would conceal this limitation.

The same-centroid contrast shows that changing the explicit scoring offset is sufficient to produce this cooler-head change within the fixed program and data. It does not identify a hydraulic causal mechanism or establish that prior removal will help a new population. The nearest-neighbor full-source tie with the much simpler prior-neutral centroid also prevents attributing that particular gain to a new nonparametric representation.

Predictor	Selection mean mask error	Evaluation mean mask error	Evaluation full-source error	Full-source vector error
Prior centroid	0.555147	0.723872	0.723872	1.000000
Prior-neutral centroid	0.348499	0.546096	0.473872	0.928741
1NN, selection chosen	0.339308	0.523901	0.473872	0.928741

Table 13: Each evaluation number uses the same 421 exposed cycles. The 24 evaluation model-mask groups contain 10104 prediction records, not 10104 independent samples.

19.3 Preserved v19: Source-mask errors are not monotone

The selected nearest-neighbor predictor makes the information issue concrete. FS1 alone has component error 0.477435, but PS1 plus FS1 has error 0.574228. TS1 alone, FS1 plus TS1, and all three sources have the same 0.473872 component error. A larger observed mask changes the neighbor geometry; it does not give this fixed predictor an option to ignore every harmful added dimension.

This observation concerns the implemented fixed decision rule. Additional coordinates alter its nearest-neighbor geometry, while the program has no learned option to ignore every harmful added coordinate. It is therefore inappropriate to impose a monotonic-error constraint on this table or treat the all-source reference as an empirical upper bound on accuracy. Figure 2 displays the complete three-by-eight comparison on one common error scale.

The FS1-only prediction also trades head errors: cooler 398, valve 153, pump 51, accumulator 202. The TS1-only errors are 0, 273, 246, and 279. Similar component means can represent very different diagnostic failures. Equal component weights are a declared experimental choice, not an elicited industrial utility function. An application with safety-critical pump decisions might rank these policies differently.

Disclosed channels	Prior centroid error	Prior-neutral error	1NN error	1NN vector error
None	0.723872	0.723872	0.723872	1.000000
PS1	0.723872	0.498219	0.520190	0.928741
FS1	0.723872	0.696556	0.477435	0.969121
PS1 + FS1	0.723872	0.550475	0.574228	0.928741
TS1	0.723872	0.478029	0.473872	0.928741
PS1 + TS1	0.723872	0.473872	0.473872	0.928741
FS1 + TS1	0.723872	0.473872	0.473872	0.928741
PS1 + FS1 + TS1	0.723872	0.473872	0.473872	0.928741

Table 14: Fixed-mask development errors, not an exhaustive industrial sensor benchmark or native-method leaderboard.

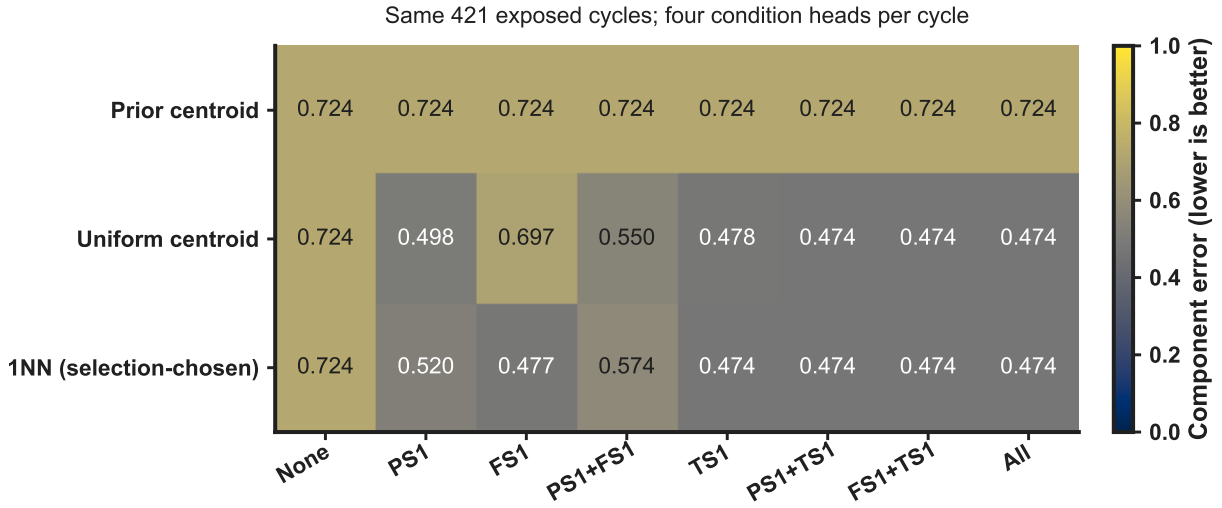


Figure 2: Mean four-head component error for three predictors and eight source masks on the same 421 exposed evaluation cycles. The 1NN model was chosen using the separate 204-cycle selection interval. Adding PS1 to FS1 worsens this fixed predictor; the full-source 0.25 gain affects only cooler and still leaves complete-vector error 0.928741.

19.4 Preserved v19: Price-sensitive controls expose a selection-transfer failure

Under the raw-sample quote, the selection-locked subset chooses TS1 for every nonzero declared budget. At budget 6660 it attains the same exposed evaluation component error, 0.473872, as the three-source reference, with quote 60 instead of 6660. At budget 660, cheapest and pressure-first acquire FS1 plus TS1 for quote 660 and again have the same error. This is a useful strong-simple-baseline observation; it is not a measured physical saving or proof that temperature alone is sufficient for future diagnosis.

Under equal source quotes, several source masks have identical selection error. The declared lower-mask-index tie rule selects PS1. At budget 1, the selection-locked policy has evaluation error 0.520190, worse than cheapest-first FS1 at 0.477435. At budget 3 it remains on PS1 while the full-source baseline has error 0.473872; even the penalized objective favors full-source on that exposed interval. We retain the unfavorable result and do not replace the tie rule with an evaluation-informed preference for TS1.

The comparison illustrates two different uncertainties. Price assumptions change feasibility and which tied validation choice is favored. Separately, equal selection scores need not transfer equally to a later condition regime. No novel optimizer is required to observe either issue. A complex adaptive policy should be compared against these fixed subset and prior-neutral baselines, with the same predictor rights and development resources, before a policy-specific contribution is claimed.

Figure 3 puts this price reversal beside the support limitation. Its panels use distinct quote units rather than a shared numerical cost axis. The policy table contains 36 summaries obtained by lookup in the frozen selected-model predictions; it adds no independent physical trial or new prediction call.

Price system / budget	Policy	Chosen mask	Quote	Evaluation component error
Raw / 6660	Stop	None	0	0.723872
Raw / 6660	Cheapest or pressure-first	All three	6660	0.473872
Raw / 6660	Selection-locked subset	TS1	60	0.473872
Equal / 1	Cheapest-first	FS1	1	0.477435
Equal / 1	Selection-locked subset	PS1	1	0.520190
Equal / 3	All-source fixed order	All three	3	0.473872
Equal / 3	Selection-locked subset	PS1	1	0.520190

Table 15: All policies share the selection-chosen 1NN predictor. Quotes from different rows' price systems are not interchangeable. The policy rows reuse fixed-mask predictions.

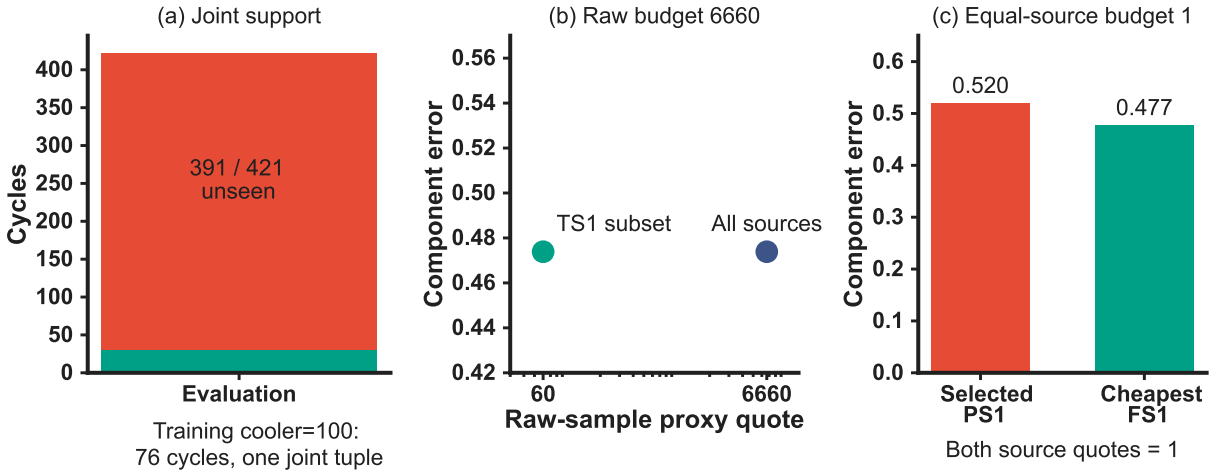


Figure 3: Joint support and two matched-price contrasts. The 76 training cooler=100 cycles have one tuple; 391 of 421 evaluation cycles have train-unseen joint conditions. At raw budget 6660 TS1 and all-source component errors tie at quotes 60 and 6660. At equal budget 1 the selected PS1 subset is worse than cheapest-first FS1 at the same quote 1. Quotes are replay proxies, not measured physical savings.

19.5 Preserved v19: V14: objective changes move errors between heads

Component and vector rules select the same subset in all nine contracts. This is a retained negative result, not evidence that their losses are generally equivalent. Every nonempty mask has the same selection vector error 100/204. Changing to that tied criterion leaves the existing quote/index preference intact. In contrast, pump changes seven masks relative to component and worst-head changes two.

At raw budget 6660, component/vector choose TS1 at quote 60; worst-head chooses PS1 at quote 6000; pump chooses FS1 at quote 600. Relative to TS1, FS1 reduces pump errors from 246 to 51 out of 421 but raises vector errors from 391 to 408 and worst-head error from 0.662708 to 0.945368. Its aggregate component error rises slightly from 0.473872 to 0.477435. A near-tie in the four-head mean therefore hides a large redistribution of diagnostic errors. Figure 4 exposes that redistribution using the same per-head denominator throughout.

Pump-focused selection has zero same-loss hindsight regret in these nine exposed contracts, including a zero-quote stop at raw budget 60 where affordable TS1 changes no pump prediction. That result concerns this narrow priority, not complete diagnosis, a generalization certificate or real pump safety. At equal budget 1 the pump rule selects FS1 with quote 1 and pump error 51/421; component/ vector/worst choose PS1 with the same quote and pump error 246/421. The negative vector and worst-head trades remain in the same table rather than being relegated to an excluded denominator.

Rule at raw B=6660	Mask/quote	Component	Vector	Worst head	Pump
Component or vector	TS1 /60	0.473872	0.928741	0.662708	0.584323
Worst-head	PS1 /6000	0.520190	0.928741	0.693587	0.584323
Pump	FS1 /600	0.477435	0.969121	0.945368	0.121140

Table 16: One selected predictor, one 421-cycle exposed cohort and one raw-price system. Each column is compared within its own loss; cross-column differences are not improvements. Masks are selection-locked, not evaluation-chosen.

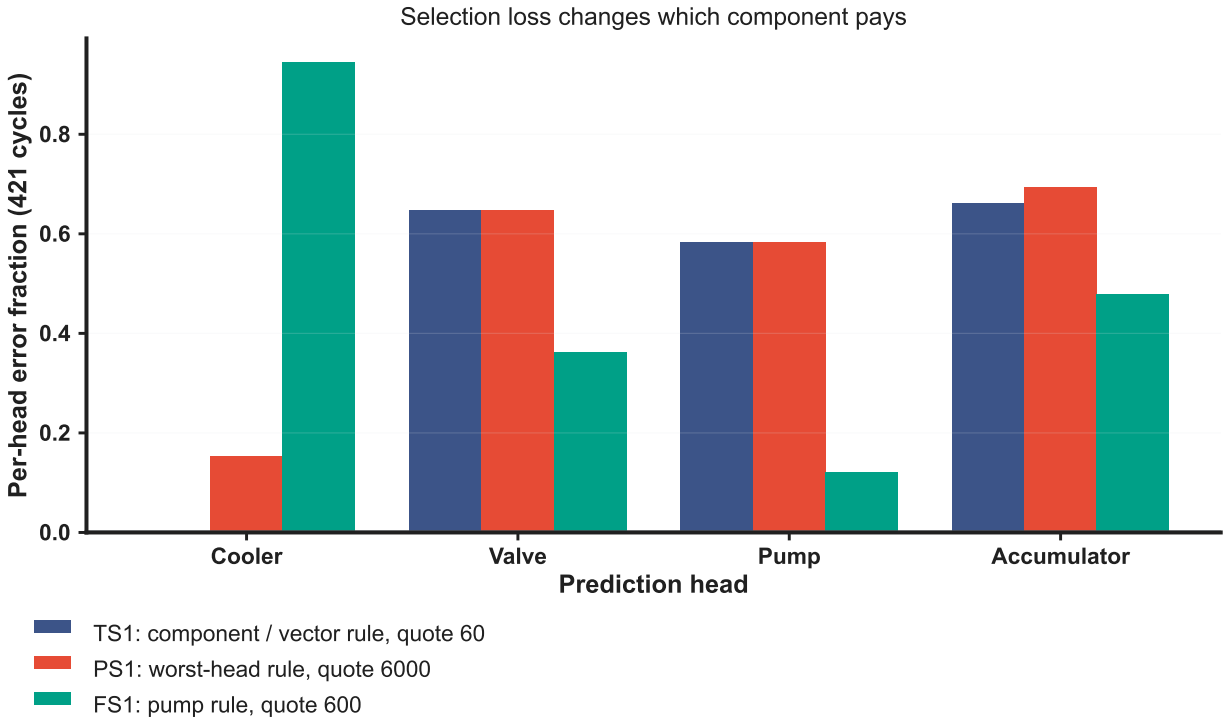


Figure 4: Per-head errors for the three masks selected by the four objective rules at raw budget 6660. Each bar divides one head's errors by 421 cycles. Pump-specific improvement trades against other heads; the worst-head rule selects a more expensive mask with worse exposed worst-head error. Fixed predictions and logical quotes, not adaptive or physical trials.

19.6 Preserved v19: V14: hindsight regret and projected dominance

Selection minimizes a development criterion; it need not minimize that criterion later. At raw budgets 6000 and 6660 the worst-head rule picks PS1 with worst-head error 0.693587, worse than TS1 at 0.662708 despite quotes 6000 versus 60. Its same-loss penalized regret is 0.075473. In this example TS1 is no worse in any of the four reported errors and costs less, so the failure is not explained by an omitted pump preference.

At equal budgets 1–3, the component rule's PS1 has same-loss hindsight regret 0.046318 against TS1 at the same quote 1. This develops, rather than erases, the earlier transfer failure. The vector rule uses the same PS1 but has zero vector regret because PS1 and TS1 tie on vector error and cost. A single chosen action can therefore be suboptimal for component error while optimal in a tied vector criterion. Comparing one rule's pump regret with another rule's component regret would silently change the estimand.

Seventeen of 36 objective-contract rows are strictly dominated in the component/vector/quote projection. The rows repeat policies across budgets and are not 17 independent failures. The projection also marks pump-focused FS1 dominated by TS1 at raw budgets where both are feasible. That diagnosis must not be promoted to universal inferiority: pump error is an omitted coordinate and is lower for FS1. At positive evaluation budgets, the projected frontier contains the empty mask and TS1; the empty mask remains because its quote is zero. Frontiers are descriptive finite-record comparisons, not a population Pareto guarantee.

19.7 Preserved v19: V14: a tight cohort ceiling and its assumptions

The new trace audit checks 5000 selected-predictor records across selection and evaluation masks. All 4375 nonempty-mask records return a training tuple;625 empty-mask records share fallback (3,100,0,90), independently checked inside training joint support. These are cached record checks, not 5000 new predictions. Selection contains 104 train-joint-seen and 100 unseen cycles; evaluation contains 30 seen and 391 unseen cycles.

Every one of the eight masks makes a vector error on all 391 unseen evaluation tuples. This is the cohort observation. The conditional closed-output proposition explains why it occurs for a training-tuple-copy decision class. The resulting exact-vector success upper bound is 30/421, or 0.071259. TS1 and the full-source reference attain it, but the affordability condition matters: TS1 is legal only at raw budget at least 60 or equal budget at least 1. Budget zero has only the empty fallback and makes 421 vector errors, so the support bound is not achieved there.

Among the 30 seen evaluation cycles, TS1 makes zero vector errors whereas FS1 makes 17. Both make 391 errors on the unseen stratum, giving total vector errors 391 versus 408. The same unseen stratum still distinguishes component priorities: pump errors are 246/391 for TS1 and 51/391 for FS1. Exact-vector loss is saturated at 1 there while head-level information remains useful. Figure 5 separates the stratum denominators; it does not provide true-label stratum information to a policy.

The ceiling says neither that unsupported hydraulic conditions are intrinsically unpredictable nor that extra information is useless. An independent-head predictor can recombine marginal classes into a tuple outside training support; a different decoder may do so as well. Such systems violate the proposition's restricted-output premise and require their own matched evaluation. Actual adaptive-policy value, abstention coverage and independent physical transfer remain unmeasured.

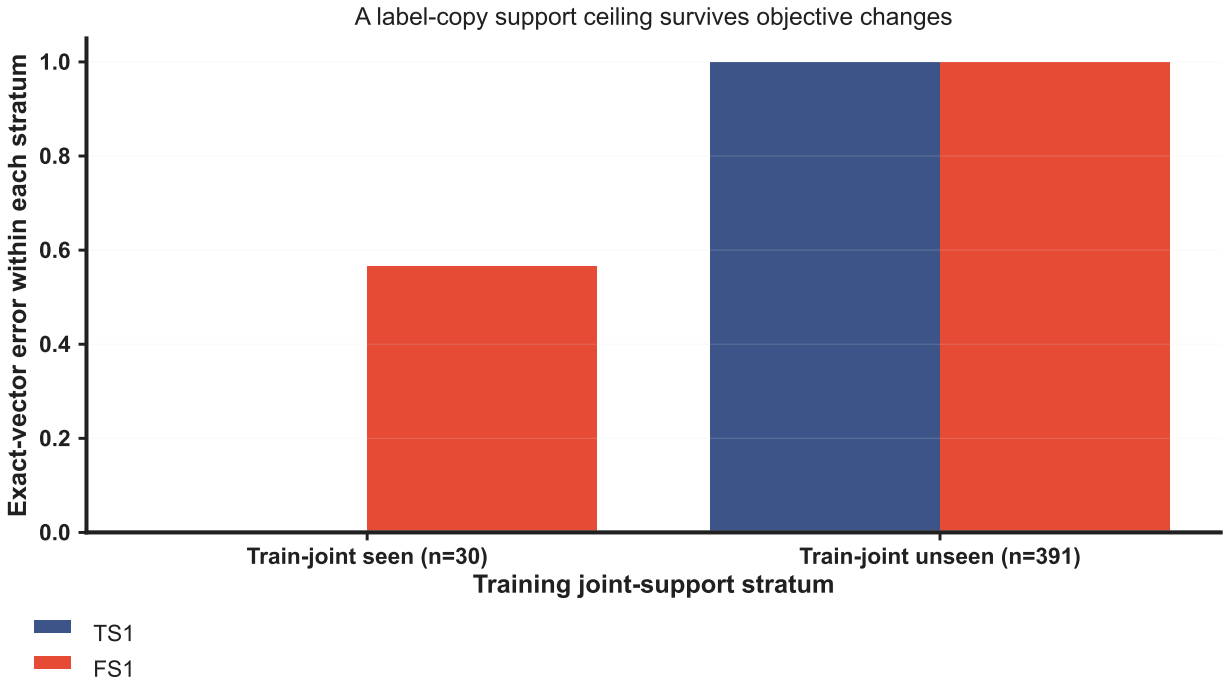


Figure 5: Retrospective exact-vector error on 30 train-joint-seen and 391 train-joint-unseen evaluation cycles. All eight masks fail every unseen tuple under the checked output-class restriction; only two masks are shown. The bound 30/421 is an upper bound, attained by TS1 only when affordable. It does not apply automatically to independent-head predictors or changed decoders, and zero-coverage abstention is not exact success.

20 Preserved v19: matched compositional neighbor voting

The closed-output bound is conditional on predicting a training tuple. To test a broader decision class, we reuse the exact v12 standardized training bank and rank neighbors using squared Euclidean distance over disclosed dimensions only. For each head and candidate label, the score is the number of its labels among the k neighbors divided by that label's training count. Maximization is separate for each head, with smaller numeric labels breaking exact score ties. Different heads can therefore choose labels supported by different neighbors and compose a tuple never observed jointly during training.

The five neighborhood sizes are fixed to k=1,3,5,11,31 before executing the run. Class counts, the 1,540 training targets, the nine standardization means/scales and training coordinates come from the byte-frozen v12 bank. We estimate no new standardizer, optimize no parameter and add no input. The operation is new local inference with an initialized voting baseline, not five newly fitted neural models. These ranking scores are not calibrated probabilities and are not used to declare selective coverage.

A source mask exposes exactly its three per-channel statistics. No unrequested coordinate, profile label, stable flag, cycle index or condition-coded stratum enters the distance or vote. The empty mask uses the original headwise training mode for all k. The train, selection and evaluation intervals remain 1..1540,1561..1764 and 1785..2205. Twenty-cycle chronological gaps are preserved without treating them as an independence guarantee.

One k is selected by the total component error on 204 selection cycles averaged equally over all eight masks, with smaller k on ties. The selected k and exact selection summaries are written before the current evaluation feature pass. Researchers had already exposed this evaluation interval in

earlier versions; the procedural ordering does not restore blindness. The selection criterion remains an exploratory development rule whose correspondence to the nine actual acquisition budgets must itself be examined later.

The run checks exact equality between k=1 outputs and all 5,000 prior selected-1NN records across 625 cycles and eight masks, including 625 empty-mask outputs. This is stronger than comparing aggregate errors. A sampled second implementation recomputes 210 nonempty-mask predictions from raw six-cycle features with standard-library sums, exact rational vote comparisons and independent sorting. All 25,000 cached prediction records are also independently rescored.

Execution accounting distinguishes reusable fitting from new predictions. We initialize one class-balanced voting baseline, estimate zero new parameters, compute 4,375 nonempty-mask query-distance arrays, use 625 empty-mask fallbacks and emit 25,000 records across the five configurations. A subsequent support audit reads those records without making new predictions. None of these repeated masks/configurations creates additional scientific units, physical rigs or independent final cycles.

$$v_{h,l}(x,k,m) = \frac{\sum_{j \in N_k(x,m)} I\big(y_{j,h} = l\big)}{n_{\text{train},h,l}} \tag{21}$$

$$\hat{y}_h = \arg\max_l v_{h,l}(x,k,m) \tag{22}$$

$$k^\star = \arg\min_k \sum_{m=0}^{7} \sum_{i \in S_{\text{selection}}} \sum_{h=1}^{4} I\big(\hat{y}_{i,h,k,m} \neq y_{i,h}\big) \tag{23}$$

20.1 Preserved v19: The selection-locked answer remains negative

This subsection preserves the v15 all-mask selection contract. V16 below changes the selection objective, rather than rewriting the original negative result.

Selection summed component error counts over the eight masks are 2215,2218,2225,2223 and 2216 for k=1,3,5,11,31 respectively. Each total has 6528 head-mask-cycle comparisons, reusing 204 underlying cycles. Thus k=1 wins, and even the nearest k=31 alternative is one comparison worse under this declared criterion. That one-count gap is not a stability certificate or a population significance result.

Because the admitted configuration is exactly k=1, all 3,368 evaluation cycle-mask records match the original baseline. Every same-cycle component comparison is tied. The selection-locked mask rule under the same 0.05 normalized quote coefficient also preserves all nine price-budget decisions: raw budgets at least 60 choose TS1, equal budgets at least 1 choose PS1, and zero budgets use no source. No observed admitted policy gain can be claimed.

The already exposed evaluation grid has lower summed component counts for several unselected configurations:7058,6955,6871,6840 and 6817 in the same k order, over 13472 head-mask-cycle comparisons. Choosing k=31 from that table would use evaluation labels. We report the divergence as a development diagnostic, not as the result of the selection procedure or independent tuning evidence. The selected policy remains k=1 even though retrospective results would tempt a different choice.

At full source mask, every configured k makes the same head error counts (0, 273, 246, 279) and 391 vector errors among 421 cycles. Therefore simply enlarging the local neighborhood does not repair the full-source representation in this grid. More available statistics do not guarantee that this fixed distance metric retrieves a useful condition mixture. The training bank's 76 cooler 100 examples all share joint tuple(100,100,0,90), which is a concrete conditional-support limitation rather than a missing marginal cooler class.

k	Selection component errors /6528	Exposed eval errors /13472	Selected?
1	2215	7058	Yes
3	2218	6955	No
5	2225	6871	No
11	2223	6840	No
31	2216	6817	No

Table 17: Eight-mask totals use the same 204 selection or 421 exposed evaluation cycles repeatedly. Only selection chooses k; evaluation columns are development diagnostics, not alternative admissible winners.

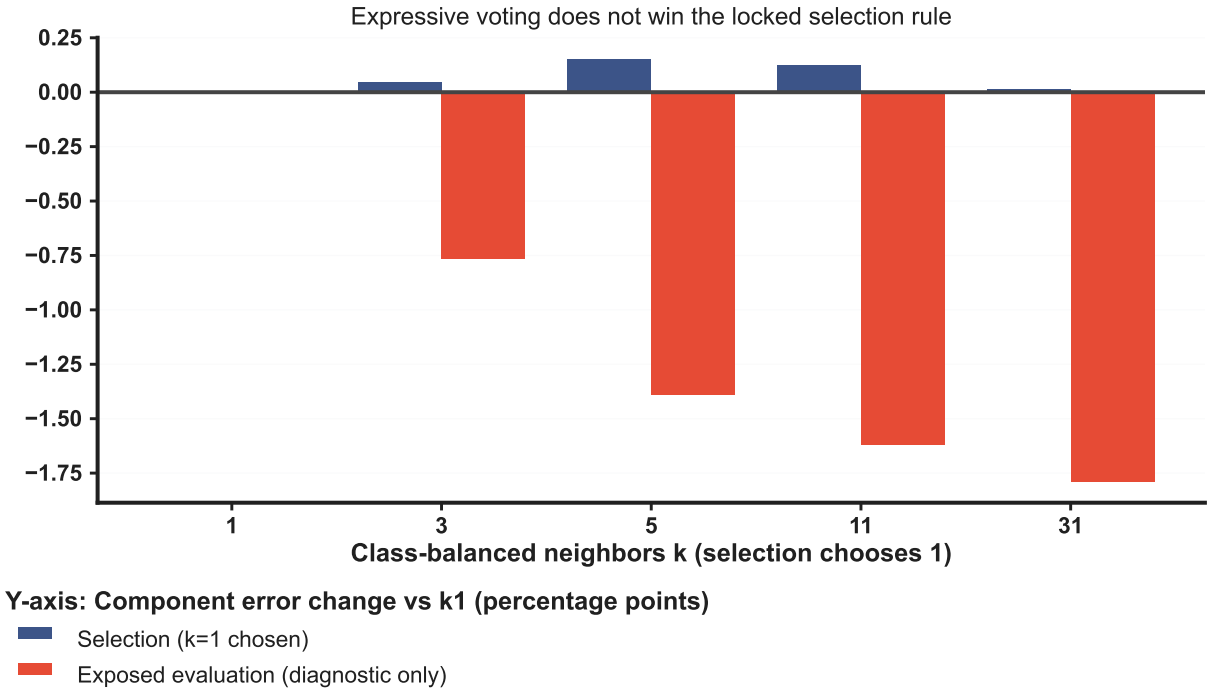


Figure 6: New v15 fixed k grid. Changes in equal-mask component error are measured relative to k=1 within each partition. The selected k remains 1 despite better exposed evaluation totals for unselected variants. Underlying denominators are 204 and 421 cycles, reused over 8 masks and 4 heads; these are not independent samples or final evaluation gains.

20.2 Preserved v19: A class-conditional ceiling can be escaped without winning the experiment

The first completed run stored retrospective support summaries only for the selected k. To avoid hiding behavior of the predefined broader decision class, we then freeze a separate explicitly post-hoc support-audit plan. It reads all 25,000 predictions and reports 160 partition/k/mask/seen-unseen stratum rows. No k is added, selected k is not changed, no new model is fit and no new predictions are generated. Its labels and train-support strata remain unavailable to the policy.

The unselected k=3 model with PS1+FS1, mask 3, outputs training-unseen tuples on 32 of the 391 training-unseen evaluation cycles and gets ten of those complete tuples correct. Its total vector errors fall from 391 under k=1 to 381, giving 40/421 complete successes. This exceeds the earlier joint-copy ceiling 30/421 because the new decoder violates the premise of that proposition. It is a constructive counterexample to a universal-impossibility reading, not a newly selected deployable winner.

Other variants expose a different tradeoff. For FS1 only, k=11 correctly predicts 14 training-unseen tuples but makes 392 total vector errors, while original k=1 has 408. These correct novel outputs demonstrate expressibility without proving broad diagnosis. At k=3 PS1+FS1 the chosen cost quote would be 6600 raw-sample units or 2 equal-source units, exceeding raw budget 6000; the favorable full-vector diagnostic must not be silently inserted into a cheaper admitted policy.

The new class-balanced decoder still cannot emit a per-head label absent from training. Its ability to output any combination of observed marginal labels says nothing about whether neighborhood geometry supports that combination, whether the acquisition mask is affordable or whether selection can identify it without evaluation feedback. In particular, all full-source configurations retain the original 391 errors, and the only admitted predictor still obeys the old training-tuple restriction on these saved outputs.

A stronger follow-up should reserve independent final condition/unit data before changing feature metrics or choosing head-specific source rights. It should compare joint copying, factorized votes and established compositional prediction under the same disclosed coordinates and complete costs, retaining cheap fixed-subset controls before any adaptive search. The present negative selection result supplies a falsifiable obstacle: a new model must improve a predeclared selection criterion and transfer without using the exposed evaluation grid to choose itself.

21 Preserved v19: selection order is a matched fixed-policy ablation

The v15 result is tied to its declared criterion: choose one k by component error summed equally over eight masks, then choose a budget-feasible mask. A k that is good on average over masks need not be best for the mask actually affordable at deployment. V16 benchmarks this staged rule against joint selection of (k,mask) from the same five-by-eight cached prediction bank. This is a predictor-selection and fixed-source control, not a newly trained adaptive acquisition agent.

Both rules have the same selection data, training bank, projected features, possible source masks and offline tuning rights. They differ in the objective used to choose the configuration. Joint selection minimizes selection component error plus lambda times the mask quote divided by the sum of the three source quotes, subject to the same hard per-cycle budget. The staged first step instead averages all masks, including masks unaffordable under a particular cap. Deterministic ties favor smaller quote, then smaller k and mask. Lambda is predeclared as 0, 0.01, 0.05, 0.2 or 1; raw and equal price books keep their original nine budget cases.

Controls include the best feasible mask with k fixed at 1, the cheapest nonempty source, a maximum-source-count affordable mask, and the exact expected loss of uniform sampling over feasible masks including empty. The label-free controls read no selection label when choosing a mask. The uniform result averages the cached outcomes analytically; it is not additional sampled trials. All controls obey the same deployment cap, although actual spent quotes may differ. Offline feature collection and tuning costs are not measured by these source quotes.

The full choice uses all 204 selection cycles. Four leave-one-contiguous-block-out views use 153 cycles to choose and 51 to audit. Three forward-prefix views use the first 51, 102 or 153 cycles and audit only the next 51. The leave-block procedure may use later selection cycles and is explicitly not a forward-deployment simulation. Full choices are sealed before this execution decodes evaluation prediction records, but all 421 evaluation outcomes were exposed in earlier work. Four evaluation blocks have 106,105,105 and 105 cycles. These dependent slices do not supply independent final data or population confidence intervals.

BRiG-AFA [3] motivates keeping the predictor fixed when isolating an acquisition-horizon effect. Our changed k therefore cannot be interpreted as a planner effect. Its reviewed unit-cost fixed-horizon setting also differs from these source quotes. Janisch et al. [20] distinguish cost-weighted and hard-budget objectives and validate model choices; we therefore report lambda and cap sensitivity separately. Neither native system is implemented, and no author-reported score is pooled with local results. A faithful native comparison still requires matched data, predictor, action granularity, information, budgets and evaluation access.

21.1 Preserved v19: A conditional predictor gain leaves diagnosis and benchmark limits intact

At the prespecified lambda 0.05, all five raw-price cases preserve the staged configuration. The three positive equal-quote caps all choose PS1 as before but replace k1 with k3. Thus three of nine configuration cases change, while zero of nine source masks change. These three caps repeat one configuration and the same 421 cycles; they are not three independent successful experiments. Joint selection is no worse on its own selection objective by candidate-set inclusion, which does not establish generalization.

For that repeated PS1 comparison, selection component errors decrease from 249 to 248 of 816 head-cycle comparisons. Exposed evaluation errors decrease from 876 to 799 of 1684, a difference of -77/1684 or -4.5724 percentage points. Head errors change from (65, 273, 246, 292) to (0, 273, 246, 280). Vector errors remain 391 of 421 for both. This refines the v15 selection-contract negative result without claiming a repaired complete diagnosis, independent final improvement, or an acquisition-mask gain.

The strong fixed-source comparison remains unfavorable to a general winner claim. The already executed TS1/k1 control has 798/1684 component errors at the same equal-source quote 1, one fewer than joint PS1/k3. Its selection count is 249/816, so the joint rule cannot use its better evaluation count to choose it. This is an explicitly retrospective diagnostic against a previously executed fixed control, not a new evaluation-tuned policy. The v15 all-mask criterion still selects k1 exactly as originally recorded.

Lambda 0 changes five full-cohort configuration cases and worsens exposed evaluation error in two raw-price cases. At raw cap 6000, selecting PS1/k3 rather than TS1/k1 lowers selection error by 1/816, increases quote from 60 to 6000, and increases evaluation error by 1/1684. At raw cap 6660, PS1+FS1/k5 lowers selection error by 5/816, increases quote from 60 to 6600, and increases evaluation error by

75/1684. A respected hard cap alone therefore does not prevent an expensive unfavorable transfer. Lambda 0.01, 0.05 and 0.2 change only the three equal-price configurations; lambda 1 changes none.

The one-count selection advantage is not stable identification of an optimal configuration. At equal cap 1 and lambda 0.05, joint selection differs from its full-cohort choice in one of four leave-block views and two of three forward-prefix views. One leave-block choice becomes FS1/k31; the first two forward prefixes retain PS1/k1. On the four exposed evaluation blocks, the locked joint-versus-staged component counts are 251 versus 282, 233 versus 244, 195 versus 205, and 120 versus 145. Vector counts remain 106,105,95 and 85 in both rules. These are dependent diagnostics of the same run, not four independent transfer replications.

Ground-truth joint tuples absent from training occur in 100/204 selection cycles and 391/421 evaluation cycles. Pump-label total variation between these intervals is 266/421; individual evaluation blocks further differ. These label diagnostics are unavailable to the chooser and are not a causal explanation of every error. They quantify why repeated chronological reuse cannot replace independently reserved conditions or physical units. The v16 study obtains no new rig, feature, prediction, parameter fit, physical cost measurement or adaptive rollout; its supported outcome is a sharper baseline and selection-scope comparison.

Fixed comparison at equal quote 1	Selection component errors /816	Exposed eval component errors /1684	Vector errors /421
Staged / fixed-k1, PS1	249	876	391
Joint (k,mask), PS1/k3	248	799	391
Existing fixed TS1/k1 diagnostic	249	798	391

Table 18: Same 204/421 cycles and equal source quote; the TS1 row is an already executed fixed-source diagnostic, not a newly selected evaluation winner. Quote equality does not establish measured physical cost equality.

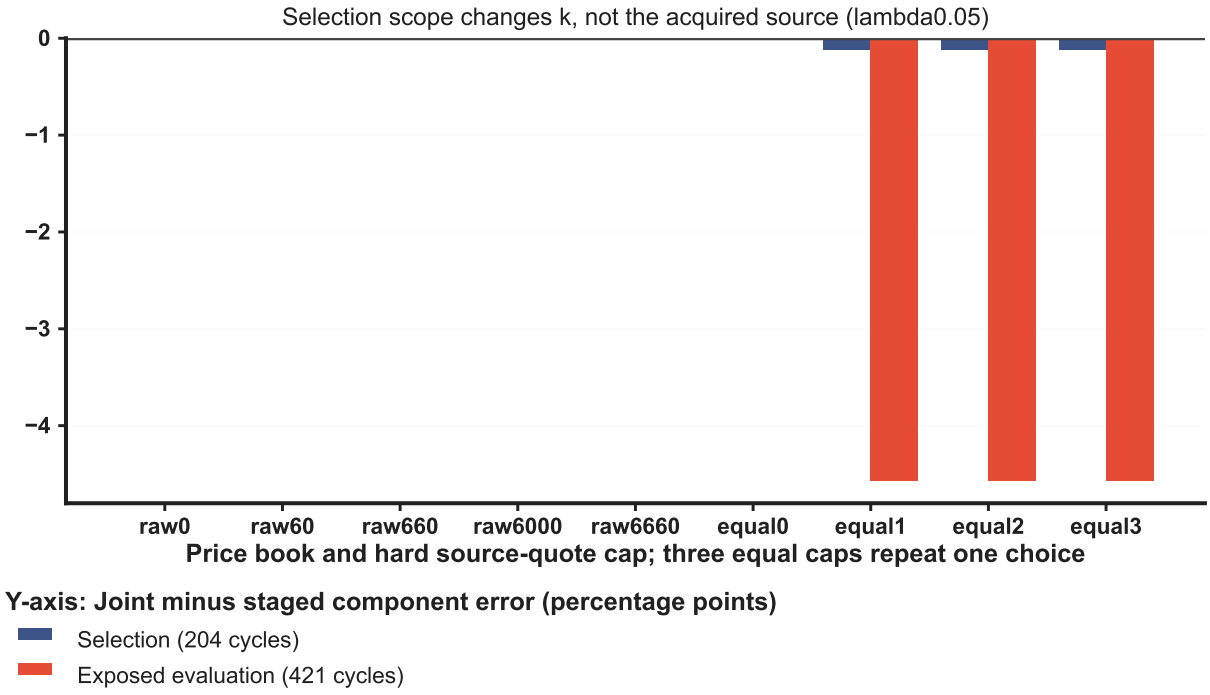


Figure 7: V16 matched selection-order benchmark at lambda 0.05. Bars show joint-minus-staged component error within each partition. The three equal-price caps repeat the same PS1/k3 versus PS1/k1 choice; all source masks are unchanged. Evaluation is previously exposed development reuse, not an independent final improvement.

22 Preserved v19: observation-dependent acquisition with a frozen terminal predictor

V17 executes a deployable-information acquisition rule rather than another fixed-mask selection audit. The terminal predictor remains the v15 k-neighbor decoder; k1 is primary and k3 is a prespecified sensitivity. Policy fit uses cycles 1561–1662 (102), policy dev uses 1663–1764 (102), and scoring retains the previously exposed 1785–2205 interval (421). The original predictor bank was fitted on 1540 prior cycles. No chronological reuse interval becomes a new independent final.

A gate first requests one root source. It receives only that source's observed complete-cycle mean, compares it with a fit-derived threshold, and either stops or requests one other source. Leaves contain the root and at most one additional source, with distinct left and right masks. Root thresholds are the sorted fit means at floor(101q), for q equal to one quarter, one half and three quarters; each branch must contain at least 16 fit cycles. Thresholds are serialized with float.hex. For each root and threshold, fit labels select the two affordable leaf masks by exact component-error plus lambda-normalized mean-quote objective. This fits an acquisition policy; it does not refit the frozen terminal predictor.

The decision function receives only the immutable policy, acquired-source means and acquired mask. A private replay service checks the hard quote cap, charges a source, then reveals its mean. The controller receives no cycle identity, target, unseen source value or cached terminal prediction. Only after STOP does the harness retrieve the frozen prediction for the terminal mask. Previously computed mean, standard deviation and slope from an acquired source remain valid terminal features; no unacquired source's features are admitted. This is an inspected trusted-code API boundary, not OS isolation.

best_fixed enumerates every feasible mask, including empty and the full three-source mask where affordable. A gate uses at most two sources. adaptive_enabled includes all those fixed candidates plus the fit-trained nonconstant gates, while gate_only probes the gate family and explicitly falls back if none is feasible. All families share k, the hard cap, lambda and the dev objective. Dev ties favor lower mean quote, then fixed over gate, then lexical policy serialization. Inclusion makes adaptive_enabled's dev objective no worse mechanically; it supplies no generalization guarantee. Equal caps do not imply equal realized mean expenditure.

The prelocked grid contains two k values, five lambdas and nine price/cap cases: 90 settings and three selected families per setting. The primary is equal-source cap 2, lambda 0.05 and k1. Raw and equal quotes remain separate proxy units. Gates are fit before dev selection; all choices are sealed before the current pass decodes evaluation means, labels or prediction records. Earlier exposure is explicit. The full grid, stronger fixed controls, permission fixtures and unfavorable outcomes are retained.

BRiG-AFA [3] supplies a methodological comparator: its fixed-predictor ablation isolates acquisition while inference uses observed values and budget rights. We do not implement its Bellman networks or native datasets. Its reviewed fixed-horizon unit costs also differ from our STOP and grouped-source quotes. Janisch et al. [20] distinguish cost-weighted objectives from budget constraints. Our chronological gate class and their hierarchical A2C are not interchangeable benchmark implementations, and external scores do not enter the local result table.

22.1 Preserved v19: Adaptation executes, but the primary selector keeps its fixed policy

At the prespecified equal cap 2, lambda 0.05 and k1, adaptive_enabled selects the same fixed PS1 mask as best_fixed. Both incur 876 of 1684 component errors and 391 of 421 vector errors on exposed evaluation, with mean quote 1. Adding the gate family therefore changes neither selected acquisition nor prediction in the primary comparison. Its dev objective ties a gate, and the predeclared simpler-policy rule preserves the fixed policy; reporting gate_only instead would change the selection contract.

The forced gate first acquires TS1 and adds PS1 below its fit threshold. Its fit, dev and evaluation terminal masks respectively use the two-source leaf in 26 of 102, zero of 102 and 22 of 421 cycles. Evaluation component errors are 798 of 1684, below the primary PS1 control. However, the preserved fixed TS1/k1 control also has 798 component errors and 391 vector errors. The gate spends 443 rather than 421 equal-quote units with no terminal error reduction. This stronger source control prevents an apparent adaptive victory. Its historical selection access differs from the current selector and is disclosed rather than used as an evaluation oracle.

Across all 90 settings, adaptive_enabled selects a gate in six; four settings reduce evaluation component error, two increase it, and 84 tie the matched fixed selector. These counts repeat one k3 gate across caps 2 and 3 and three lambdas, not six independent trials. The common gate reads PS1 and adds FS1 only above its fit median. Its two-source frequencies change from 51 of 102 fit cycles to 17 of 102 dev cycles and 97 of 421 evaluation cycles, so even one locked rule has different realized expenditures across intervals.

At k3, equal cap 2 and lambda 0 or 0.01, the fixed comparator is PS1+FS1. The gate lowers evaluation component errors from 924 to 861 and quote sum from 842 to 518, but vector errors rise from 381 to 391. Its component advantage therefore coexists with ten fewer complete diagnoses. Component and

vector denominators are respectively 1684 and 421; they are different observables, not interchangeable performance summaries. Cap 3 repeats this same contrast.

At k3 and lambda 0.05, the fixed comparator changes to PS1. The gate wins dev component counts by 245 versus 248, yet evaluation increases from 799 to 861 errors and quote sum from 421 to 518. Vector errors remain 391. Thus a three-count dev advantage produces 62 additional evaluation component errors and 97 additional proxy quote units. We retain this selection-to-evaluation reversal rather than choose lambda after viewing the exposed outcomes.

Equal-cap comparison	Dev errors /408	Eval errors /1684	Vector errors /421	Eval quote sum
Primary k1 fixed / union	249	876	391	421
Primary k1 forced gate	249	798	391	443
Historical TS1/k1 diagnostic	not a new selector	798	391	421
k3 lambda 0 fixed / gate	245 / 245	924 / 861	381 / 391	842 / 518
k3 lambda 0.05 fixed / gate	248 / 245	799 / 861	391 / 391	421 / 518

Table 19: Same exposed 421 cycles, no pooled external scores. Primary cap is 2; cap 3 repeats the k3 rows. Historical TS1 has different selection access. A shared hard cap does not match realized expenditure, and quoted units are not measured physical costs.

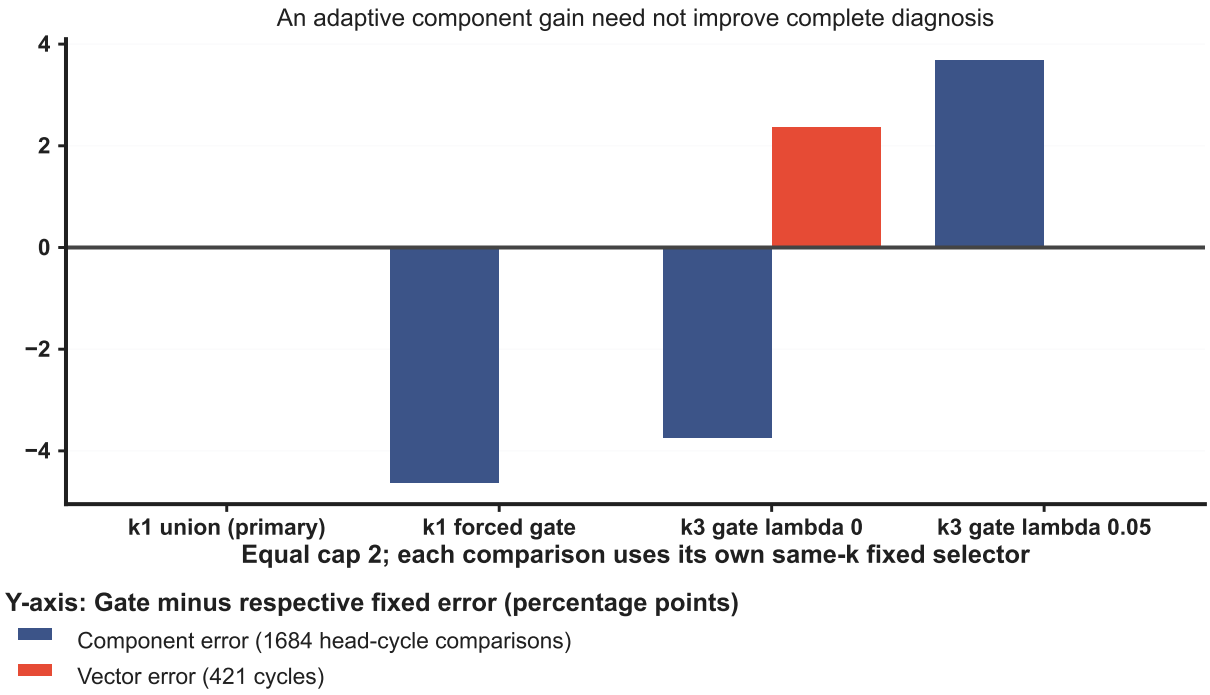


Figure 8: V17 gate-minus-respective-fixed differences on exposed evaluation. Primary union retains fixed PS1; forced k1 gate matches historical TS1 errors while spending 22 extra units. k3 lambda 0 reduces component errors but increases vector errors; lambda 0.05 worsens component errors. Component points use 1684 comparisons, vector points 421 cycles. Repeated caps are not independent trials.

22.2 Preserved v19: Policy-fit support is a new limiting assumption

A post-hoc frozen-label diagnostic reveals that all 102 policy-fit cycles share the truth tuple (100, 100, 0, 90). That tuple occurs in only 2 of 102 policy-dev cycles and 30 of 421 evaluation cycles. Thus the gate is not trained across the diverse target conditions it later encounters. This diagnostic is derived after locking the policy and never enters its decisions. We retain the predeclared chronological split; shuffling it after seeing this limitation would answer a different question.

Where fit outcomes tie, forcing different leaf masks can make cost and deterministic tie rules identify a gate without identifying condition-dependent predictive utility. The primary forced gate's acquisition branch is absent from dev and reappears on evaluation. This is an observed support and selection limitation, not proof that observation-dependent acquisition cannot work or that one distribution shift causes every error. A dedicated diverse policy-fit cohort and independent condition/rig evaluation are still required.

The new work comprises 390 fitted gate models, 780 fitted leaf actions, nine fit-derived thresholds, 790 dev candidate scores and 270 selected-policy records. Selected-policy execution produces 168750 replay trajectories, of which 35000 execute nonconstant gates. A trajectory is one policy/cycle path, with zero to three source requests, not one request or independent sample. These traces contain

130459 source requests, counted separately. Candidate-dev replay and historical diagnostic execution are separately counted. Frozen predictor parameter fits and forward calls are zero, while new policy fitting and replay are explicitly nonzero.

Recorded 60-second sensor cycles are fully observed offline. Sequentially revealing their statistics assumes the stored channels coexist and that hiding/revealing a value does not change that recorded cycle. Seeing a completed cycle's mean before requesting another channel does not establish that the same past interval can later be measured online: buffering, remeasurement and latency requirements are untested. This study does not model intervention effects, acquisition delays or online equipment control. No physical adaptive sensor experiment, independent final unit or native BRiG-AFA/A2C reproduction is conducted. An independent mathematical replay review cannot replace those missing data.

22.3 Preserved v19: Oracle-separated loss diagnostics locate the remaining acquisition gap

A new post-hoc diagnostic holds all390 fitted gate-context records, the frozen predictions, prices, hard cap, k and lambda fixed. For each cohort, A is the per-cycle all-feasible-mask oracle mean, P the current two-leaf oracle mean, G the actual gate mean and F the locked fixed-policy mean. The exact identity G-F=-(F-A)+(P-A)+(G-P) splits the difference into negative headroom, the root/leaf/cost-constraint representation gap and privileged routing regret. The last two are not pure learning errors or causal effects: the oracle observes labels and unseen predictions unavailable to an executable policy.

The primary inherited k1/equal cap2/lambda0.05 forced gate has F=13561/25260,A=10201/25260,P=12391/25260,G=12413/25260 on421 exposed evaluation cycles. Thus H=56/421,E=73/842,R=11/12630 and G-F=-287/6315. The pair oracle equals the existing TS1/k1 objective. No routing of these same two frozen leaves can strictly beat that TS1 comparison on this particular finite cohort and scalar loss. The actual gate retains798/1684 component errors and391/421 vector errors, with22 extra quotes.

A predetermined secondary k3/equal cap2/lambda0 gate gives component-error counts F924,A625,P789,G861. The exact decomposition is-299+164+72=-63 over1684 component comparisons. The gain coexists with vector error381->391 and quote842->518. We therefore preserve the objective-specific benefit and the complete-diagnosis regression instead of pooling the metrics.

The analysis produces1,170 gate/cohort records and4,680 support-stratum records on the same625 cycles, with an exact arithmetic alternative verifier and ten explicitly post-hoc illustrative cycles. It adds no policy fitting, policy selection, predictor forward calls, independent device, physical acquisition or external native-method trial. Empty support strata have null means. The392/421 primary root means outside the fit min/max range are an observed extrapolation diagnostic, not a density-support or causal test.

$$A = \left(\frac{1}{n}\right) \sum_{i=1}^{n} \min_{m \in \mathcal{F}_B} u_i(m) \tag{24}$$

$$P = \left(\frac{1}{n}\right) \sum_{i=1}^{n} \min(u_i(m_L), u_i(m_R)) \tag{25}$$

$$G - F = -(F - A) + (P - A) + (G - P) \tag{26}$$

Quantity	Primary k1 objective	Secondary k3 errors /1684
Locked fixed F	13561/25260	924
All-mask oracle A	10201/25260	625
Two-leaf oracle P	12391/25260	789
Actual gate G	12413/25260	861

Table 20: Same predictor, price, cap, lambda and finite cohort; oracles are privileged retrospective lower bounds.

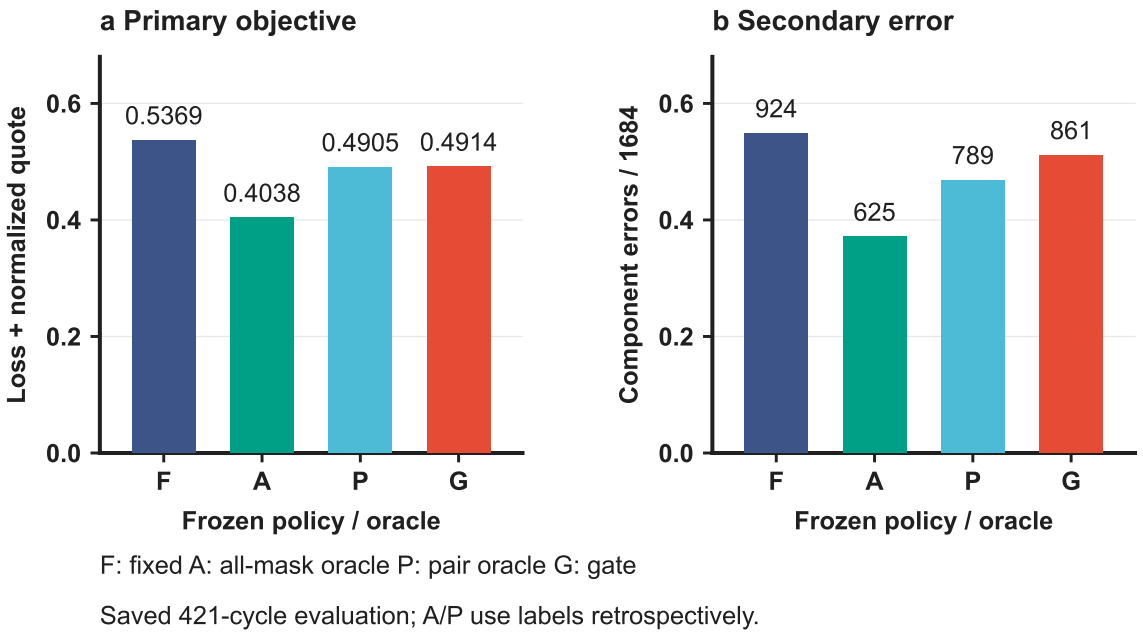


Figure 9: New v19 decomposition of already saved predictions. Left: primary scalar objective including normalized source quote. Right: secondary lambda0 component-error counts. All oracles use true labels and unobserved mask predictions only after policy locking; their gaps are not deployable gains or independent benchmark results.

22.4 Preserved v19: The next experiment must change evidence, not only threshold search

The primary pair-oracle tie implies a narrow but useful design decision: spending more search on the same two leaf masks cannot establish strict improvement over TS1 under the declared finite loss. A new leaf class or predictor is a separate intervention and must be tested with its own information and cost rights. Positive routing regret elsewhere identifies a privileged opportunity, without proving that acquired root observations can learn to exploit it.

The expanded Korean handbook distinguishes source, feature, mask, predictor, policy, fitting, dev selection, exposed evaluation, cost scalarization, hard feasibility, component/vector losses and oracle rights. It includes exact stepwise recorded cases and a visual reading map to generative surrogates, Bellman risk-to-go, hierarchical costly features, learned information-gain controls and assumption-aware offline evaluation [3,9,10,11,20,21]. These methods are primary-source comparisons rather than native executions here.

The next confirmatory study needs diverse policy-fit conditions and an independently reserved device or time regime, with measured acquisition availability, failure and latency. Complete-cycle offline means, synthetic permission fixtures and one-rig exposed replay do not establish physical adaptive sensing, native AFA superiority or confidence coverage.

23 Preserved v19: Discussion and Limitations

The point of the diagnostic ordering is to identify what an acquisition comparison could establish. A failed predictor check directs attention to scoring and support; it does not prove that a better planner is impossible. A successful predictor check only admits a policy comparison; it does not certify safe deployment. Here that distinction stops the earlier zero-coverage result from being hidden by a later component-error gain, and stops a raw-price saving from hiding an equal-price loss.

V14 adds a distinct limitation to planning escalation. A mismatch between component utility and complete-vector utility can coexist with identical selected actions, and a changed head priority can improve one component while damaging another. When legal outputs stay within training joint support, loss tuning alone cannot create a missing complete label tuple. This finite class-conditional statement is stronger than reporting a high error but narrower than declaring planning futile. It redirects the next experiment toward supported output representations and externally justified utilities, without claiming that the research workflow itself causally improves productivity.

23.1 Preserved v19: Falsification and alternative explanations

We distinguish a data-admission failure from a poor valid prediction, an empty selective denominator, a resource rejection, and an unavailable independent evaluation. The original v9 class-support failure remains visible. V11 zero coverage remains a null selective error rather than a zero error. V12 produces valid four-head outputs under all 24 masks/models, but high vector error remains a negative outcome rather than a rejected record.

The original v12 prices, candidate order, tie rule and usefulness threshold remain unchanged. V14 deliberately changes the declared decision objective in a post-hoc continuation; earlier aggregate evaluation outcomes were already known. Its four-objective plan and exact input/code snapshots were fixed before the new calculation, with no additional objective or tie rule added after observing that calculation. This protects procedural reproducibility, not independence. If the empty fallback leaves training joint support, the training-support ceiling cannot be applied unchanged. If a new predictor emits novel joint tuples, it needs a different output-class bound. If a chosen worst-head policy loses to a cheaper fixed subset, that loss remains even when another policy attains its narrow pump goal.

Our candidate explanations have explicit falsifiers. If a same-centroid prior-neutral contrast does not change predictions on new, supported conditions, the present scoring explanation does not transfer. If a stronger simple predictor with adequate condition coverage removes a planner's advantage, a policy-specific claim weakens. If correct source identity and disclosure remove a reported cost advantage, that advantage was bookkeeping or leakage. If a history-aware baseline reconstructs the relevant executor state, a richer hidden-state claim is unnecessary.

No failure in this finite development sequence establishes a population rate. Conversely, a narrow, correctly scoped descriptive report should not be rejected for lacking an experiment that it explicitly does not claim to have performed. Research Forge records the current claim, supporting artifact, contrary observation, and next required evidence separately. This claim-scope discipline prevents speculative next steps from being promoted to supported results.

23.2 Preserved v19: Boundaries and the next confirmatory study

The most important limitation is support and exposure, not manuscript length. The evaluation segment is already known, uses one rig and a small channel subset, and contains many joint conditions absent from the training prefix. We have no verified physical experiment/session/asset map and no fresh independent final set. A chronological gap cannot supply those missing identities. The 421 cycles are a descriptive denominator, not a proved independent sample count.

The feature representation is deliberately simple and uses a completed cycle. The 1NN fit changes several inductive assumptions and has no calibrated confidence. A constant-label full-source predictor can improve component mean by correcting only one head. Fixed subset baselines do not establish adaptive or nonmyopic value, and the present grouped action cannot test all of ProbeFlow's proposed partial-disclosure and shared-cache distinctions.

The next confirmatory study should first obtain reliable physical-block provenance and adequately supported condition combinations. Predictor and future-observation model development must remain within training; an explicit development/calibration/final resource envelope must precede any final evaluation. The comparator set should retain the prior-neutral centroid, fixed subset design, and history-aware simple baselines, alongside faithful native or clearly declared adapted literature methods. Computational planning costs, failures, tuning, and teacher work must be included or separately bounded.

Within that new setting, a typed executor can vary overlapping sources, partial disclosure, failed requests, and cache reuse while preserving public information and quotes across policies. A matched comparison would be needed to isolate the value of executor state from a history-aware baseline that reconstructs it. For now, neither industrial generalization, an independent algorithmic effect, nonmyopic superiority, nor a certified risk guarantee is established.

24 Preserved v19: Conclusion

ProbeFlow's inherited support-first audit separates scoring, coverage and acquisition. The new frozen-objective analysis shows that changing the loss may preserve every action, redistribute errors between heads, or select a more expensive dominated choice. A checked training-tuple output restriction also yields a tight exact-vector ceiling on this exposed cohort, with explicit fallback and affordability conditions. The conclusion is not a new adaptive optimizer's victory: it is a reproducible account of which decision changes help which loss, and which complete outputs this predictor cannot express. Independent supported data, meaningful utilities and faithful comparator implementations remain prerequisites for broader policy claims.

The v15 compositional neighbor study crosses the closed-output boundary for some unselected diagnostics but yields no admitted predictor or policy gain. Exact baseline reproduction and the preserved k=1 choice prevent expressive counterexamples from being sold as a validated selection victory. The remaining research target is a supported, independently reserved comparison with stronger representation and unchanged information rights.

V16 shows that a stronger matched selection contract can change the admitted predictor while leaving the acquisition choice and complete-vector failure unchanged. Its component gain is conditional on an exposed development interval and does not beat every same-quote fixed control. Cost-weighted counterexamples and chronological instability make independent data and fixed information rights the next admission gates.

V17 genuinely fits acquisition gates and executes observation-dependent offline trajectories. Nevertheless, the primary union retains its fixed policy, and a forced gate matches the existing TS1 control while spending more. Selected k3 gates reveal component/vector conflicts and a dev-to-evaluation reversal. A single-condition policy-fit interval further blocks transfer claims. More acquisition machinery is not established as the remedy; diverse policy fitting, measured costs and independent evaluation are the next evidence requirements.

V19 locates the primary gate limitation in its frozen action class. Under identical feasible masks, cached predictor, cost normalization and scalar loss, G-F=-H+E+R separates available headroom, the root/leaf/cost constraint bundle and a privileged-oracle routing gap. The primary two-leaf oracle already equals the historical TS1 control, so rerouting only these leaves cannot strictly improve that objective on the same exposed cohort. The secondary gate reduces component errors but increases vector errors. These finite diagnostics prioritize supported policy fitting and action representation before added planning complexity; they do not establish population impossibility or a guaranteed learnable gain.

Acknowledgements

This work was created as part of the No-More-Slops group activities at Pseudo Lab.

References

[1] N. Helwig, E. Pignanelli, and A. Schütze. Condition monitoring of hydraulic systems. UCI Machine Learning Repository, 2015.
[2] Y. Kobayashi, Z. Jing, J. Yao, H. Namkoong, and S. Joshi. Learning-To-Measure: In-Context Active Feature Acquisition. arXiv:2510.12624v2, 2026.
[3] J. Feng, Q. Li, and Y. Li. BRiG-AFA: Bellman Risk-to-Go Learning for Non-Myopic Active Feature Acquisition. arXiv:2608.02305v1, 2026.
[4] Z. Zhao, R. Xu, and Q. Ni. Learning Source Acquisition Policies by Offline Planning. arXiv:2609.14299v1, 2026.
[5] L. Aronsson and M. Haghir Chehreghani. Non-Myopic Active Feature Acquisition via Pathwise Policy Gradients. arXiv:2605.05511v2, 2026.
[6] Y. Geifman and R. El-Yaniv. Selective Classification for Deep Neural Networks. arXiv:1705.08500v2, 2017.
[7] D. Golovin and A. Krause. Adaptive Submodularity: Theory and Applications in Active Learning and Stochastic Optimization. JAIR 42; arXiv:1003.3967v5, 2011.
[8] H. Shim, S. J. Hwang, and E. Yang. Joint Active Feature Acquisition and Classification with Variable-Size Set Encoding. NeurIPS, 2018.
[9] Y. Li and J. B. Oliva. Active Feature Acquisition with Generative Surrogate Models. ICML, PMLR 139:6450-6459, 2021.
[10] I. C. Covert, W. Qiu, M. Lu, N. Y. Kim, N. J. White, and S.-I. Lee. Learning to Maximize Mutual Information for Dynamic Feature Selection. ICML, PMLR 202:6424-6447, 2023.
[11] H. von Kleist, A. Zamanian, I. Shpitser, and N. Ahmidi. Evaluation of Active Feature Acquisition Methods for Time-varying Feature Settings. JMLR 26(60):1-84, 2025.
[12] C. Ma, S. Tschiatschek, K. Palla, J. M. Hernandez-Lobato, S. Nowozin, and C. Zhang. EDDI: Efficient Dynamic Discovery of High-Value Information with Partial VAE. ICML, PMLR 97:4234-4243, 2019.
[13] L. Aronsson, A. Rahbar, and M. Haghir Chehreghani. A Survey on Active Feature Acquisition Strategies. IJCAI Survey Track, pages 7740-7748, 2026.
[14] G. C. Cawley and N. L. C. Talbot. On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation. JMLR 11(70):2079-2107, 2010.
[15] S. Varma and R. Simon. Bias in error estimation when using cross-validation for model selection. BMC Bioinformatics 7:91, 2006.
[16] D. R. Roberts et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography 40:913-929, 2017.
[17] Y. Geifman and R. El-Yaniv. SelectiveNet: A Deep Neural Network with an Integrated Reject Option. ICML, PMLR 97:2151-2159, 2019.
[18] R. El-Yaniv and Y. Wiener. On the Foundations of Noise-free Selective Classification. JMLR 11(53):1605-1641, 2010.
[19] K. Dembczyński, W. Waegeman, W. Cheng, and E. Hüllermeier. On label dependence and loss minimization in multi-label classification. Machine Learning 88:5–45, 2012.
[20] J. Janisch, T. Pevny, and V. Lisy. Classification with costly features in hierarchical deep sets. Machine Learning 113:4487-4522, 2024.

[21] J. Janisch, T. Pevny, and V. Lisy. Classification with Costly Features Using Deep Reinforcement Learning. AAAI33(01):3959-3966,2019.

A Computation and exact-byte reproducibility

The bounded v12 run took approximately 20.22 seconds on the local host, including parsing, fitting, selection, evaluation, and output generation. Shared fitting took about 0.0067 seconds. Across the eight evaluation masks, measured inference totals were about 0.165 seconds for the prior centroid, 0.141 seconds for the prior-neutral centroid, and 8.970 seconds for 1NN. Full-source inference alone was about 0.030, 0.026, and 1.954 seconds, respectively. These are single sequential wall-time observations, not randomized throughput comparisons; no hardware efficiency claim is made.

The historical v12 run uses only the Python standard library and existing local data. It makes zero model API calls, network requests, or downloads. This does not mean that no predictive model was fitted: centroids, scalers, and a nearest-neighbor reference bank are fitted machine-learning components. It means that no external or local generative-model endpoint was invoked for the experiment.

PLAN.json, raw-file SHA-256 values, the inherited feature extractor hash, exact consumed runner bytes, frozen fitted parameters, selection predictions, frozen model/policy choice, evaluation predictions, event ordering, and result metrics are saved in a fresh run directory. The exact code snapshot is created before fitting so later source changes cannot silently substitute for the consumed implementation. Hashes establish byte identity, not dataset truth, independent review, or scientific validity.

Eight focused checks cover train-only fitting, exclusion of unqueried dimensions, the isolated prior-term counterfactual, preservation of the original predictor, deterministic model selection and nearest-neighbor ties, quote/budget separation, and explicit nonempty metric denominators. A separate saved-trace audit recomputes counts, losses, class support, and policy costs without new prediction calls.

A v11 audit discovered that two aggregate ledger fields had added unlike price units. We retained that original report, marked those scalar totals invalid, and recomputed a scenario-separated ledger from its unchanged prediction trace. Its per-arm costs and predictive outcomes were unaffected. The exact consumed code was archived before correcting the future runner. This incident motivates a practical invariant: numerical provenance must bind units as well as hashes. A matching hash cannot repair a dimensionally invalid sum.

B Artifact map and execution recipe

The historical v12 execution project is projects/research_manuscripts_v12_20261018/probeflow. Its PLAN.json and run_experiment.py define the bounded study; test_experiment.py checks contracts; SOURCES.json records inherited primary-source review scopes; README_KO.md explains the result in Korean; and paper.json is the structured source of that historical baseline manuscript. The run directory is runs/research_manuscripts_v12_20261018_01/probeflow.

The run contains locked_plan.json, source_receipt.json, consumed_code/run_experiment.py, frozen_fit.json, selection_predictions.jsonl, frozen_selection_and_policies.json, evaluation_predictions.jsonl, results.json, and events.json. The current-run selection artifact precedes current evaluation decoding in the event log. Neither that log nor a hash erases the previous exposure disclosed in the plan.

From the repository root, run Python with UTF-8 and bytecode disabled: python -X utf8 -B projects/research_manuscripts_v12_20261018/probeflow/test_experiment.py. To repeat the sensor computation, pass a new, unused output directory to run_experiment.py –out. Never rerun into an existing result directory. The original v9/v11 assets and active research skills are preserved.

Policy rows are lookups into the selected predictor's saved mask predictions: 36 budget-price-policy summaries do not constitute 36 independent tests. There are 4896 selection prediction records and 10104 evaluation records, over 204 and 421 unique cycles respectively. The audit should check per-head errors, full-vector errors, duplicate cycles within an arm, permitted source dimensions, exact fit/selection hashes, and price units before consuming a summary.

C Admission checklist for a future policy claim

- Dataset identity: pin source bytes, extractor version, units, target definitions, and physical block provenance.
- Support: disclose marginal labels and joint/condition support separately; retain unsupported cases in the declared denominator.
- Predictor: demonstrate useful complete and partial-input behavior before attributing value to policy search; retain strong simple alternatives.
- Information rights: bind public observations, private cached work, permitted future forecasts, and target exclusion for every action.

- Prices: separate hard caps from soft penalties; charge a shared source once; never add unlike units or label replay quotes physical savings.
- Selection: freeze candidate set, mask weighting, tie rules, thresholds, and tuning budget; label exposed development and independent final evidence correctly.
- Comparison: distinguish native reproduction, substantive common-contract adaptation, and hand-crafted reference policies.
- Reporting: preserve empty selective denominators, failed arms, unfavorable simple baselines, exact consumed code, and subsequent corrections.

D Primary-source review boundaries

The v13 source ledger records a new direct review of selected primary HTML pages. Some entries cover official abstracts only; others cover identified definitions and method sections. Such a review verifies a bounded description, not complete proofs, code identity, native accuracy, or the correctness of every author claim. No external benchmark result becomes a local observation.

SOURCES.json binds each review note to its URL, scope, review time, and a canonical review-record SHA-256. It does not invent a publisher-byte hash: original response bytes were not archived, and their hash is null. Local experiment and code hashes refer to actual files. CLAIMS.json links empirical statements to these immutable records and separates observed results from proposals and untested generalization.

E V14 decision-audit provenance and verification

The new analysis is preserved in runs/research_advancement_v14_20261019_01/probeflow. PLAN.json and locked_plan.json record four objectives, both price systems, access rights, tie rules and post-hoc scope. admission.json pins five input snapshots and the exact consumed runner before its subprocess executes. frozen_objective_decisions.json precedes current evaluation aggregation; manifest.json seals 11 artifacts. No original v11/v12 execution was overwritten. The profile labels and prediction banks are preserved bytes, not newly acquired data.

run_experiment.py performs the new selection/objective/Pareto/stratum calculation. verify.py independently implements count and rational-choice arithmetic within the same author lane, reconciles 16 selected-predictor partition-mask groups, checks 5000 label-copy records, recomputes 36 decisions and 144 same-loss hindsight comparisons, and checks the empty fallback separately. Eight small authored boundary tests cover loss ranking reversal, empty denominators, pump indexing, hard budgets and price units, exact ties, Pareto-equivalent points, modal-fallback nonclosure and a finite support ceiling. Fixtures and record checks are not extra sensor experiments or external scientific review.

The run result SHA-256 is 47e689aba91964956f7f41fdc69934978851beaefcb0ab25e71c32aa0fb23f83. The consumed code SHA-256 is d9b3d27819ed6969233ce7f94e30c8e3684c6dbbe1b9b0425ac3cae9ee3373f0. In that inherited support/objective execution, new fits, predictions, adaptive rollouts, model API calls and physical trials are all zero. The exact input manifest and scopes belong to the reproduction record; their hashes do not establish blindness, physical independence or signed endorsement.

F V15 compositional prediction and support-audit inventory

The current structured manuscript and Korean guide are projects/research_advancement_v15_20261019/probeflow/paper.json and EXPLAINER_KO.md. PLAN.json and analyze.py define the five-config local voting experiment. The consumed snapshot is runs/research_advancement_v15_20261019_01/probeflow/analysis_01, containing exact code/plan, raw and fitted-bank copies, input_lock.json, frozen_selection.json,25000 new prediction records, results.json and receipt.json. This implementation uses the already installed NumPy 2.3.2 plus Python standard library; no dependency was added.

Reproduction uses a fresh path: python projects/research_advancement_v15_20261019/probeflow/analyze.py –out runs/research_advancement_v15_20261019_01/probeflow/NEW_RUN. SUPPORT_PLAN.json and support_audit.py define a separate post-hoc cached-record diagnostic in support_audit_01; it produces 160 support-stratum rows with 0 new predictions. verify.py independently checks all cached metrics and 210 sampled neighbor predictions. Fits 0, API calls 0 and new prediction records 25000 are distinct accounting fields.

G V16 matched-policy benchmark inventory

The new source is projects/research_advancement_v16_20261019/probeflow; execution is runs/research_advancement_v16_20261019_01/probeflow/analysis_01. PLAN.json, analyze.py, verify.py

and six input snapshots are fixed by input_lock.json before selection. policy_locks.json records 2160 choices before this execution decodes evaluation predictions. Historical evaluation exposure remains explicit. No prior source or execution is overwritten.

policy_results.jsonl contains 2160 fit/test metric pairs. joint_sequential_contrasts.jsonl contains 360 paired rule contrasts; evaluation_block_diagnostics.jsonl contains 1080 dependent block rows; selection_stability.jsonl preserves 270 choice-sensitivity records; label_shift_diagnostics.json reports six exposed-label cohorts. These rows repeatedly reuse 625 cycles from one rig and are not independent sample counts. The uniform feasible-mask control uses exact rational expectations rather than sampled random runs.

verify.py imports no analysis functions. It independently reaggregates all 25000 cached records, recomputes 2160 selections and their fit/test metric pairs, and checks 1080 evaluation-block metrics. Exact Fraction strings preserve expected uniform-control counts and all objective ties. This alternative implementation validates finite local arithmetic, not external scientific replication. ACCOUNTING.json and CLAIMS.json preserve the zero new-prediction/fit/API counts and limits on transfer and cost interpretation.

The v16 primary-source review reopens BRiG-AFA v1 HTML and the official Janisch2024 publisher HTML. Review locators and short excerpt hashes identify the local reading records, not publisher response bytes or external certification. BRiG-AFA, hierarchical CwCF, native adaptive acquisition and their published performance results are not implemented or pooled into the local benchmark.

H V17 policy execution and evidence inventory

The v17 source and execution roots are projects/research_advancement_v17_20261019/probeflow and runs/research_advancement_v17_20261019_01/probeflow/analysis_01. input_lock.json binds consumed PLAN.json, analyze.py, verify.py and eight byte-preserved input snapshots before fitting. fitted_gates.json records the 390 branch fits; candidate_dev_scores.json records 790 candidate scores; policy_locks.json fixes 270 selected records before this pass reads evaluation outcomes.

rollout_traces.jsonl records 168750 selected-policy/cycle trajectories, including requests in actual order, revealed means in hexadecimal notation, quote, terminal mask and selected cached prediction. policy_results.json separates fit, dev and evaluation metrics; paired_contrasts.json retains all 90 settings. Permission poisoning executes the actual controller on 1560 synthetic boundary fixtures; these are not independent real-data tests. ACCOUNTING.json reports request events separately from trajectories, candidate-dev replay, historical controls and zero predictor refits.

verify.py imports no analyze functions. It rederives raw source means, fit thresholds, best leaf assignments, dev winners, 810 cohort metric groups and all 168750 traces. This alternative author implementation and subsequent nonauthor review concern finite arithmetic and inspected information rights; neither certifies population validity. AUTHOR_DIAGNOSTICS.json is a separately marked post-hoc support/request-count derivation with source/profile/trace/code hashes. It never changes the locked policy or split.

I Source inspection boundaries

[1] Direct v13 primary HTML review: dataset, variables, citation, and license; physical independent-block IDs and deployment costs unverified.

[2] Direct primary HTML review of setup and predictor/acquisition methods in Sections 3-4.2; no native implementation, full proof, or numerical reproduction.

[3] Direct primary HTML review of fixed-predictor risk-to-go and reviewed unit-cost budget setup, Sections 3-4.2; no native reproduction. V16 re-opened primary HTML Sections3-5 and7; same-predictor controls, unit costs and native-comparison limits only.

[4] Direct primary HTML review of grouped-source setup and method, Sections 2-3.4; no native teacher/scorer training or benchmark reproduction.

[5] Direct primary HTML review of setup and pathwise-policy approach; no complete gradient proof or native reproduction.

[6] Direct v13 review of risk/coverage definitions and SGR assumptions; no risk guarantee is transferred to the present descriptive threshold.

[7] Direct v13 review of definitions and a greedy-guarantee statement only; no industrial, history-dependent-price, or current-predictor theorem claimed.

[8] Official abstract: joint acquisition/classification, cost-sensitive stop-or-acquire agent and set representation only.

[9] Official abstract: generative-surrogate support for model-based acquisition; no architecture/proof/ benchmark verification.
[10] Official abstract: amortized conditional mutual information and learned greedy selection; no numerical superiority transferred.
[11] Official abstract: acquisition-distribution and offline/semi-offline evaluation assumptions only; no guarantee applied here.
[12] Official abstract: partial VAE and information-gain acquisition; no local implementation or external benchmark reproduction.
[13] Official proceedings abstract: POMDP framing and model-based/model-free/hybrid taxonomy only.
[14] Official abstract: finite-sample selection-criterion overfitting and evaluation bias; no quantitative bound applied.
[15] Direct primary HTML abstract/background: selection inside evaluation and nested validation; no claim that random nested CV solves this temporal problem.
[16] Direct primary HTML review of structured dependence, blocked evaluation, and extrapolation tradeoffs; no independence guarantee for a fixed gap.
[17] Official abstract: integrated prediction/rejection learning; not implemented here.
[18] Official abstract: foundational risk/coverage and reject-option framing; noise-free assumptions are not asserted for Hydraulic.
[19] Direct primary publisher HTML inspection of metadata, abstract and selected Sections 4–5 and 7.5: loss-specific Hamming/subset-zero-one decisions and tradeoffs. Binary multilabel theory is not transferred as a theorem for these multiclass heads; no proof/code/benchmark reproduction.
[20] V16 direct publisher HTML review of Sections3.1 and6.2, cost-assignment discussion and limitations. Objective/cap distinctions and validation protocol only; no hierarchical A2C implementation or numerical reproduction.
[21] Official publisher abstract and bibliography plus author implementation pointer checked2026-10-19. Cost-sensitive acquire-or-classify framing only; no complete proof/code audit or native benchmark reproduction.